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Project report on

**“IN-PIPE INSPECTION ROBOT USING MACHINE LEARNING FOR
CORROSION AND CRACK DETECTION”**

Submitted in partial fulfilment of the requirement for the award of

MASTER OF TECHNOLOGY

In

ROBOTICS AND ARTIFICIAL INTELLIGENCE

Submitted by

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CERTIFICATE

This is to certify that the Project entitled "**In-Pipe Inspection Robot Using Machine Learning for Corrosion and Crack Detection**" is carried out by **PRAJWAL H U (1MS23RAI09)**, student of **RAMAIAH INSTITUTE OF TECHNOLOGY**, Bangalore-54, during the academic year **2024-2025** and is for partial fulfilment for award of degree in Master of Technology in Robotics and AI. It is attested that the report contains all corrections and suggestions made during the assessment. Furthermore, it is attested that none of their work has ever been submitted to a university for the purpose of receiving a degree, diploma, or certificate, even one with a similar designation. The project report has been approved since it complies with the project-related academic requirement for the aforementioned Master of Technology in Robotics and Artificial Intelligence.

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DECLARATION

I, **PRAJWAL H U**, bearing **USN: 1MS23RAI09**, a M.Tech in Robotics and AI student at Ramaiah Institute of Technology in Bengaluru, hereby declare that the project work included in the dissertation I submitted to Visvesvaraya Technological University in the academic year 2024–2025 under the title **“In-Pipe Inspection Robot Using Machine Learning for Corrosion and Crack Detection”**, is an accurate representation of the submission of this project satisfies a portion of the criteria for the Master of Technology in Robotics and Artificial Intelligence degree. This project work is done under the guidance of **Dr. Girish V. Kulkarni**, Assistant professor, Department of Mechanical Engineering, Ramaiah institute of Technology, Bangalore. Additionally, no one else has ever submitted the outcome represented in the project work report for the award of any degree.

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ABSTRACT

Efficient pipeline maintenance is crucial for ensuring the safety and reliability of critical infrastructure in sectors such as oil and gas, water distribution, and sewage management. This project presents the development of a smart in-pipe inspection robot designed to detect defects like cracks and corrosion within pipelines. The system is built around a Raspberry Pi, which serves as the central processing unit for controlling hardware components and executing on-board algorithms. A high-resolution camera captures real-time visual data, which is analyzed using the YOLOv5 object detection model to identify surface defects. By integrating robotics, machine learning, and embedded systems, the solution offers a portable, cost-effective alternative to manual inspection. The robot minimizes human exposure to hazardous environments while enabling timely and accurate defect detection. This work marks a significant step toward scalable, autonomous infrastructure monitoring and highlights the potential of AI-powered embedded robotics in predictive maintenance applications.

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CHAPTER 1

INTRODUCTION

Pipelines play a pivotal role in modern infrastructure, serving as the backbone for industries such as oil and gas, water distribution, and sewage management. These critical systems enable the efficient and safe transport of essential resources over vast distances, contributing to the functioning of economies and communities. However, the structural integrity of pipelines is constantly challenged by environmental factors, aging, and operational stress, making regular inspection and maintenance imperative to prevent failures that can lead to catastrophic environmental, financial, and safety consequences.

The traditional methods of pipeline inspection often involve manual efforts, which are not only time-consuming but also prone to errors and inefficiencies. These conventional approaches are limited in their ability to detect minor but critical defects, particularly in inaccessible or hazardous environments. The growing demand for more reliable, efficient, and automated inspection techniques highlights the need for innovative solutions.

This report introduces the concept of In-Pipe Inspection Robotics, a cutting-edge approach that leverages advanced robotics and machine learning technologies. These systems are designed to autonomously navigate pipelines, identify structural defects such as cracks and corrosion, and provide real-time data for precise maintenance planning. By integrating robotics and artificial intelligence, in-pipe inspection robots offer a transformative solution to enhance the efficiency, safety, and reliability of pipeline maintenance operations.

1.1. Background

Pipelines play a pivotal role in the functioning of critical infrastructure across various industries. They facilitate the transportation of resources like oil, gas, water, and sewage, supporting economic activities and ensuring public health and safety. However, over time, pipelines are exposed to environmental factors, operational stresses, and aging, which can result in the formation of defects, such as cracks, corrosion, and blockages. These issues not only compromise the structural integrity of pipelines but also pose significant risks to the environment, public safety, and financial investments. Therefore, regular pipeline inspection and maintenance are essential to prevent catastrophic failures and ensure the continued safe operation of these systems.

1.2. Traditional Pipeline Inspection Methods

Historically, pipeline inspection has been carried out using manual inspection techniques or semi-automated methods. Manual inspections involve human inspectors physically entering the pipeline to check for defects. While effective, these methods are labor-intensive, time-consuming, and expose workers to hazardous conditions, such as toxic gases or confined spaces. Additionally, human error can result in missed defects or inaccurate assessments.

Pigging (Pipeline Inspection Gauge) is another method used to inspect pipelines. Pigging tools are inserted into the pipeline to assess its interior, often using techniques such as ultrasonic testing or magnetic flux leakage (MFL). While pigging is useful for inspecting long stretches of pipelines, it requires the pipeline to be shut down, leading to operational downtime. Furthermore, pigging is less effective in small-diameter pipelines or complex pipeline geometries, making it unsuitable for certain inspection scenarios.

1.3. Limitations of Traditional Methods

Despite the advancements in inspection tools, traditional methods face significant limitations:

1. Labor-intensive processes that increase costs and time spent on inspections.
2. Increased human exposure to hazardous environments, which poses safety risks.
3. Limited adaptability to complex pipeline structures and the ability to detect minor, but critical, defects.
4. Operational downtime due to pigging or manual inspections, which affects the continuous operation of the pipeline.

1.4. Advancements in Robotic and AI Technologies

Recent developments in robotics, artificial intelligence (AI), and sensor technologies have enabled more effective and automated solutions for pipeline inspection. Robotic systems, equipped with advanced machine learning algorithms and sensors, can autonomously navigate pipelines, detect defects, and provide accurate data for analysis. These systems can be deployed in confined spaces and hazardous environments, minimizing the need for human involvement and improving overall safety.

Among the various AI techniques used for defect detection, YOLOv5 (You Only Look Once, version 5) stands out as an optimal solution for real-time object detection. YOLOv5 is known for its speed, accuracy, and ability to detect multiple objects in a single pass, making it ideal for identifying and locating defects such as cracks and corrosion in pipelines.

1.5. Role of Machine Learning in Pipeline Inspection

Machine learning (ML) has revolutionized the field of automated defect detection, enabling high-speed and accurate analysis of pipeline conditions. Unlike traditional rule-based detection methods, ML models can learn from vast amounts of image data, allowing them to recognize complex patterns of defects with greater precision.

The YOLOv5 (You Only Look Once, version 5) algorithm is a state-of-the-art deep learning model widely used for real-time object detection. It processes images in a single forward pass, making it highly efficient for applications that require rapid decision-making, such as pipeline inspection. By training YOLOv5 on a dataset of pipeline images annotated with defects, the model can accurately identify and classify cracks and corrosion in real time.

The advantages of using YOLOv5 in this project include:

1. High-speed detection – Suitable for real-time processing on embedded devices.
2. Accurate defect classification – Differentiates between various pipeline anomalies.
3. Lightweight model – Can be deployed on Raspberry Pi microcontrollers and edge computing devices.

CHAPTER 2

LITERATURE REVIEW

This chapter presents a thematic review of prior research in in-pipe inspection robotics, categorizing significant contributions across robotic locomotion, machine learning-based defect detection, sensing and localization technologies, swarm systems, edge computing, and domain-specific deployments, thereby identifying research gaps addressed by the present work.

2.1. Robotic Locomotion Mechanisms

The design of locomotion systems is crucial for the robot's ability to navigate confined pipelines. Researchers have proposed diverse mobility architectures to ensure adaptability, stability, and propulsion within varying pipe geometries.

Zholtayev et al. [1] developed the Smart Pipe Inspection Robot (SIPIR) with in-chassis actuation and SLAM-based localization, achieving 88.28% detection accuracy across different pipe diameters. Reiss et al. [6] presented a modular self-propelled robot integrating ultrasonic testing and visual inspection tools for scalable pipeline maintenance. Pulles et al. [16] focused on gearbox-driven locomotion for long-distance traversal and improved stability.

Inchworm-style mechanisms have also shown promise. Xie et al. [19] designed a cam-linkage-based inchworm robot that reduces actuator complexity while enhancing traction and bidirectional motion. Qiao et al. [26] improved upon this by introducing a self-locking mechanism capable of producing 15.2 N of traction force, overcoming static friction challenges in traditional designs.

Highly flexible systems have also been developed for hazardous or complex environments. Martín-Barrio et al. [21] proposed a hyper-redundant robot with 14 degrees of freedom, supported by remote actuation and mixed-reality teleoperation. Nassiraei et al. [28] created KANTARO, a fully autonomous sewer robot that employs passive-active intelligent motion to navigate bends and junctions. Additionally, Elankavi et al. [18] designed a wheeled robot with optimized wheel angles validated via simulations and a 3D-printed prototype, enabling continuous wall contact for improved maneuverability.

2.2. Deep Learning for Defect Detection

The application of machine learning, particularly deep learning, has led to a surge in defect detection accuracy and speed in pipeline inspections. YOLO (You Only Look Once), especially YOLOv5, is widely adopted for real-time object detection due to its speed and precision.

Zeng et al. [2] enhanced YOLOv5 by integrating ECBAM and SAC modules, improving feature extraction and yielding a 5.27% increase in mean average precision (mAP). Fu et al. [7] employed image preprocessing techniques such as gamma correction and dual filtering to refine YOLOv5's performance on PE gas pipelines, achieving a 1.33% gain in mAP and 3.83% in recall.

Chen et al. [3] proposed a hybrid deep learning framework combining YOLOv5 with Vision Transformers (ViT) to analyze Magnetic Flux Leakage (MFL) signals, leading to higher precision and F1 scores. Wang et al. [13] introduced YOLO-DD, integrating RDAT, IGFS, and SE modules into the YOLOv5 pipeline, resulting in an accuracy of 82.41% at 38.25 FPS.

GAN-based augmentation techniques were explored by Chen et al. [5], who combined CycleGAN with YOLOv5 and transformer attention for real-time defect detection in petrochemical pipelines, achieving 93.10% precision and 0.92 F1-score. Vriesman et al. [14] replaced handcrafted features with Texture CNNs (TCNNs), reaching 99.20% accuracy in corrosion classification.

Liu et al. [10] implemented zero-shot learning by integrating GLIP with YOLOv5, improving real-time adaptability in water pipelines. Lightweight models were proposed by Wang et al. [9], who developed a fast and efficient YOLOv5 variant with knowledge distillation, achieving 80.5% mAP at 75 FPS—ideal for embedded systems in sewer pipelines.

2.3. Swarm-Based Robotic Inspection

To enable scalable inspection of large pipeline networks, swarm robotics offers significant advantages. Kandil et al. [4] introduced PLIERS, an IoT-enabled swarm system that leverages CNNs and ultrasonic sensors for autonomous crack detection. The modular design and cloud-based analytics achieved a high F1 score of 96%, demonstrating both precision and robustness in distributed environments.

2.4. Sensing and Localization Technologies

Reliable sensing and localization are critical for mapping pipeline defects and ensuring accurate positioning of the inspection robot.

Daniyan et al. [11] developed an inline robot equipped with ultrasonic and color sensors for defect detection and predictive maintenance in the oil and gas industry. Mekid et al. [12] used electromagnetic wave propagation modeling in PVC pipes, demonstrating effective wireless leak detection within the 100–200 MHz frequency range.

To overcome GPS-denied environments, Dong et al. [22] proposed a dual-permanent-magnet mechanical antenna system, capable of data transmission through steel pipelines up to 3 meters. Kazeminasab et al. [17] developed a modular robot with bidirectional wireless communication synchronized with a motion control algorithm for efficient operation in water distribution networks.

For underwater pipelines, Sang and Norris [20] fused sonar and visual data using a fuzzy logic-based system, achieving an RMS trajectory error of 0.481 m in GPS-denied scenarios. Unnikrishnan et al. [29] introduced a laser-based non-contact navigation system that enabled Autonomous Underwater Vehicles (AUVs) to maintain precise positioning within submerged pipelines.

2.5. Sewer and Urban Infrastructure Applications

Urban pipeline systems require specialized inspection tools due to their complex geometries and frequent blockages. Huynh et al. [27] addressed this by developing a stereo vision-based system for 3D defect reconstruction using dual HD cameras, enabling accurate detection of cracks and obstructions.

Alejo et al. [31] enhanced robotic localization using RGB-D visual odometry, Monte Carlo localization, and CNN-based manhole detection. By incorporating GIS data, they achieved a median localization error of approximately 1 meter in sewer networks—significantly outperforming traditional SLAM methods.

2.6. Edge AI and Real-Time Deployment

Deploying AI models at the edge, particularly on embedded devices, has become crucial for real-time analysis and autonomous operation.

Toma et al. [25] implemented edge ML for robotic inspection using bipartite graph matching and visual computing, showing significant improvements in processing speed and detection accuracy. Heidarian et al. [23] optimized wireless power transmission (WPT) through resonant magnetic induction to enhance the performance of subsea robots, ensuring reliable duplex communication and energy efficiency.

2.7. Tabulated Review of Literature

To provide a consolidated view of the existing research, the key contributions across the reviewed literature are summarized in Table 2.1. The studies have been categorized based on their thematic focus, including robotic locomotion, machine learning-based defect detection, sensing and localization, swarm robotics, edge computing, and application-specific innovations. This tabular representation highlights the diversity of approaches adopted by researchers, the models or mechanisms employed, and their corresponding outcomes.

Table 2.1: Thematic Classification of Literature on In-Pipe Inspection Robots

Theme	Authors	Methodology / Contribution	Key Features / Results
Robotic Locomotion and Design	Zholtayev et al. [1]	SIPIR robot with in-chassis motor actuation, SLAM	Adaptive to pipe diameters, 88.28% detection accuracy
	Reiss et al. [6]	Modular self-propelled robot	Visual/UT inspection + internal machining
	Pulles et al. [16]	Long-range gearbox-driven robot	Stable operation in tough pipeline conditions
	Martín-Barrio et al. [21]	Hyper-redundant robot with mixed-reality teleoperation	14 DOF, enhanced spatial navigation
	Nassiraei et al. [28]	Autonomous KANTARO robot	Passive-active intelligent motion for

			bends/junctions
	Elankavi et al. [18]	Wheeled robot with optimized wheel angles	Smooth wall contact, ADAMS simulation validated
	Qiao et al. [26]	Inchworm-style robot with self-locking clamps	Improved traction (15.2 N)
	Xie et al. [19]	Cam-linkage inchworm robot	Bidirectional movement, 160 N payload
ML for Defect Detection	Zeng et al. [2]	YOLOv5 + ECBAM + SAC	+5.27% mAP improvement
	Fu et al. [7]	YOLOv5 + image preprocessing (Gamma, dual filtering)	1.33% mAP, 3.83% recall
	Chen et al. [3]	YOLOv5 + Vision Transformer (ViT)	High precision & recall on MFL data
	Wang et al. [13]	YOLO-DD: YOLOv5 + RDAT, IGFS, SE modules	82.41% accuracy @ 38.25 FPS
	Liu et al. [10]	GLIP + YOLOv5 for zero-shot defect detection	Effective for real-time urban monitoring
	Chen et al. [5]	CycleGAN + YOLOv5 + transformer attention	93.10% precision, 0.92 F1-score
	Vriesman et al. [14]	Texture CNN (TCNN)	99.20% corrosion detection accuracy

	Hosseinzadeh et al. [30]	Image stitching with AI post-processing	± 0.5 mm pose estimation accuracy
Swarm-based Systems	Kandil et al. [4]	CNN-based robot swarm (PLIERS)	IoT-enabled, 96% F1-score
Sensing & Localization	Daniyan et al. [11]	Inline robot with ultrasonic & color sensors	Crack/corrosion detection validated
	Mekid et al. [12]	Wireless leak detection using EM modeling	100–200 MHz propagation in PVC pipes
	Dong et al. [22]	Dual-permanent-magnet antenna	10 m in-air, 3 m through-steel data transmission
	Kazeminasab et al. [17]	Wireless comm. + motion sync	Effective in water networks with bends
	Sang & Norris [20]	Vision-sonar fusion via fuzzy logic (AUV)	RMS error 0.481 m underwater
	Unnikrishnan et al. [29]	Non-contact laser-sectioning system	Maintains AUV center inside pipe
Sewer & Urban Inspection	Wang et al. [9]	Lightweight YOLOv5 + knowledge distillation	80.5% mAP, 75 FPS sewer defect detection
	Huynh et al. [27]	Stereo vision with 3D reconstruction	High accuracy crack classification
	Alejo et al. [31]	RGB-D + CNN manhole detection + GIS	1m localization error without GPS
Edge AI / Power	Toma et al.	Edge ML with graph-matching for	Fast real-time detection

Efficiency	[25]	IoT/UAVs	on embedded systems
	Heidarian et al. [23]	Duplex wireless power via magnetic resonance	Efficient power transfer in subsea robots

2.8. Summary

The reviewed literature highlights significant advancements in smart pipeline inspection technologies integrating robotics, AI, and advanced sensing. Researchers have focused on improving defect detection, localization, and navigation in complex pipeline environments. Various deep learning models—especially enhanced YOLOv5 architectures integrated with Transformer attention, Vision Transformers (ViT), and hybrid CNN frameworks—have substantially boosted detection accuracy, precision, and recall. Edge computing and image preprocessing techniques have further improved real-time performance. Robotic systems have evolved from single-unit designs to modular, inchworm-style, and swarm-based robots with superior traction, flexibility, and adaptability in pipelines of varying sizes and materials. Innovations include stereo vision, 3D reconstruction, ultrasonic testing, electromagnetic leak detection, and underwater inspection using AUVs. Efforts to improve wireless communication, power transmission, and GPS-denied localization using visual odometry and AI-driven methods have also shown promising results. Collectively, these innovations present a robust, scalable, and intelligent framework for automated pipeline inspection across industrial, urban, and underwater environments.

CHAPTER 3

PROBLEM STATEMENT AND OBJECTIVES

This chapter defines the core problem addressed in this project which is the need for an intelligent and cost-effective solution for detecting defects within pipelines. It also outlines the key objectives formulated to design and implement a smart in pipe inspection robot using AI and embedded systems.

3.1. Problem Statement

Current pipeline inspection methods are labor-intensive, hazardous, and inefficient, often failing to detect critical defects like cracks and corrosion. There is a need for an automated and cost-effective solution capable of real-time defect detection, precise localization, and safe operation in confined and hazardous environments. This project aims to develop an In-Pipe Inspection Robot that integrates robotics, machine learning, and sensor fusion to address these challenges.

3.2. Objectives

- Design a robot capable of navigating through pipeline while providing real-time monitoring of their internal conditions.
- Integrate machine learning model like YOLOv5 to accurately detect and identify defects such as cracks and corrosion.
- Utilize a Python-based flask application with YOLOv5 for efficient real-time image processing and defect detection.

CHAPTER 4

METHODOLOGY AND SPECIFICATIONS

This chapter explains the working method followed in building the smart in pipe inspection robot. It includes the hardware and software components used in the system along with their specifications.

4.1. Methodology

The methodology for the development of the In-Pipe Inspection Robot follows a structured approach encompassing design, hardware selection, software development, AI integration, and testing.

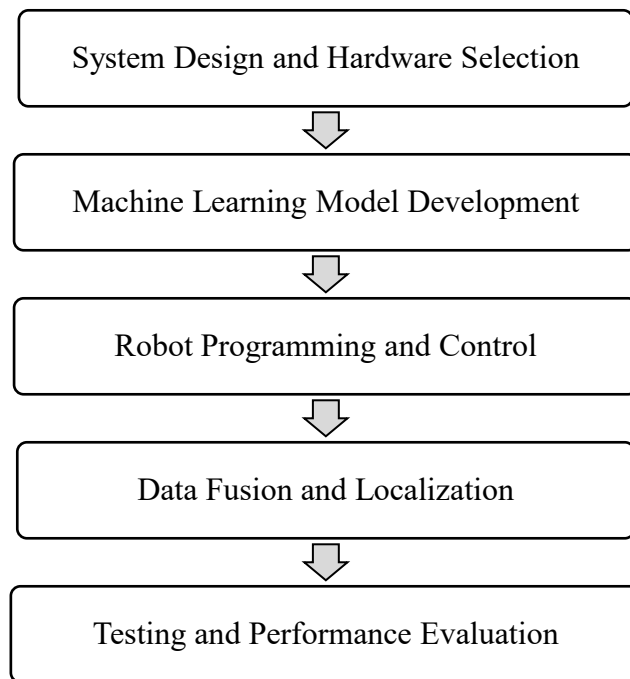


Figure 4.1: Methodology Workflow for the In-Pipe Inspection Robot.

The process is divided into the following key phases:

1. System Design and Hardware Selection

- Mechanical Structure:
 - a. The robot features a compact wheeled chassis capable of navigating inside pipelines with a fixed diameter.

- b. A mounting system is designed for the camera and sensors to maintain proper alignment for accurate image capture.
- Hardware Components:
 - a. Raspberry Pi (Model 5) – Main controller for image capture, processing, and movement control.
 - b. Raspberry Pi Camera – Used for capturing pipeline interior images for defect analysis.
 - c. Motor Driver – Interfaces the Pi with DC motors for movement.
 - d. DC Motors – Provide controlled movement.

2. Machine Learning Model Development

- Dataset Collection & Annotation:
 - a. Images of pipeline defects (focus on cracks and corrosion) are collected from real/simulated environments or open datasets.
 - b. Annotation is done using Roboflow to label cracks and corrosion areas.
- Model Training:
 - a. YOLOv5 is trained using the annotated dataset.
 - b. Roboflow is used for data augmentation (rotation, brightness, noise, etc.) and model export in PyTorch format.
 - c. Final model is optimized to run on the Raspberry Pi.

3. Robot Programming and Control

- Motor and Movement Control:
 - a. Raspberry Pi controls the DC motors via GPIO and the motor driver.
 - b. Movement routines (forward, stop, rotate) are written in Python using RPi.GPIO libraries.
- Image Processing and Detection:
 - a. A Python script captures frames from the camera using OpenCV.

- b. The YOLOv5 model is loaded and used for real-time inference on each frame to detect cracks/corrosion.
- 4. Data Fusion and Localization
 - Defect Localization:
 - a. As the robot moves, camera detects the defects and sends image to help estimate the position of detected defects.
- 5. Testing and Performance Evaluation
 - Controlled Pipeline Testing:
 - a. The robot is tested in mock pipeline segment.
 - b. Tests evaluate navigation, defect detection accuracy, and real-time response.
 - Performance Metrics:
 - a. Precision & Recall – Measure detection accuracy of cracks and corrosion.
 - b. mAP (mean Average Precision) – Evaluates overall model performance.
 - c. Inference Time on Raspberry Pi – Ensures real-time detection feasibility

4.2. Hardware Architecture

The robot is equipped with embedded hardware components that allow it to navigate pipelines and detect defects in real-time. The selected hardware components are:

1. MECHANICAL STRUCTURE

The mechanical design of the In-Pipe Inspection Robot is tailored to operate efficiently inside pipelines of fixed diameters (approximately 200 mm). The robot adopts a wheeled mechanism supported by a symmetric and lightweight frame to ensure balance and stability within confined cylindrical environments. It utilizes three spring-loaded arms, each equipped with a DC motor and a wheel, which press against the pipe wall to ensure traction and support while navigating through the pipeline.

The central structure includes a shaft that supports the entire frame and also acts as the mounting base for a high-resolution Pi camera module. This camera is

strategically positioned to maintain a fixed orientation for image acquisition and accurate defect detection using the YOLOv5 algorithm.

The robot's compact profile and symmetrical layout make it highly suitable for in-pipe operations where maneuverability and balance are crucial.

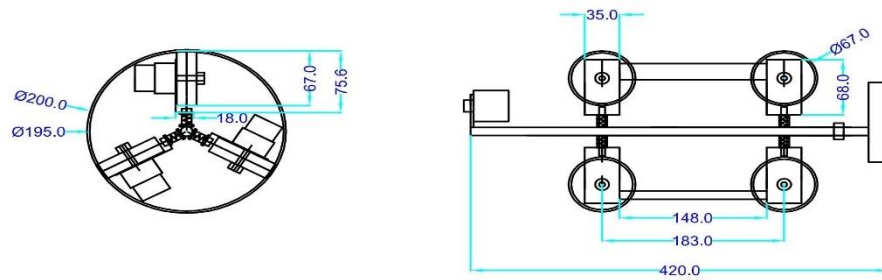


Figure 4.2: 2D CAD Design of the Robot Showing Dimensional Constraints

This top and side view illustrates the robot's design optimized for 200 mm diameter pipelines with labeled dimensions for body, motor placement, and wheel spacing.

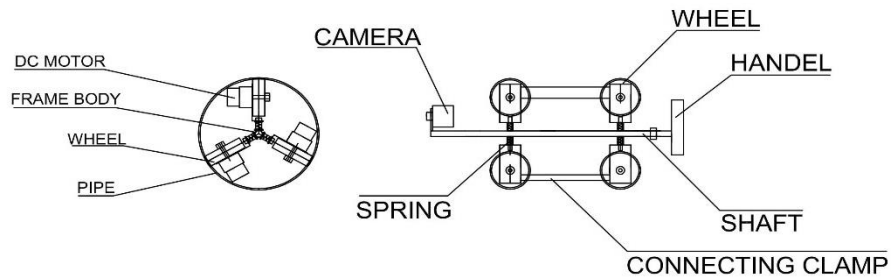


Figure 4.3: Labeled CAD Model of the Robot with Component Annotations

The figure 4.3 shows all major parts including wheels, motors, camera mount, springs, shaft, and frame body.

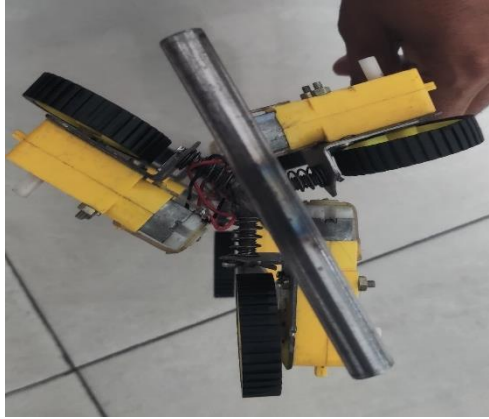


Figure 4.4: Top View of Assembled Prototype with Spring-Loaded Arms and Motors

This prototype image shows the real assembled model with motors, wheels, and camera mount integrated on spring-based arms for stable pipe wall contact.

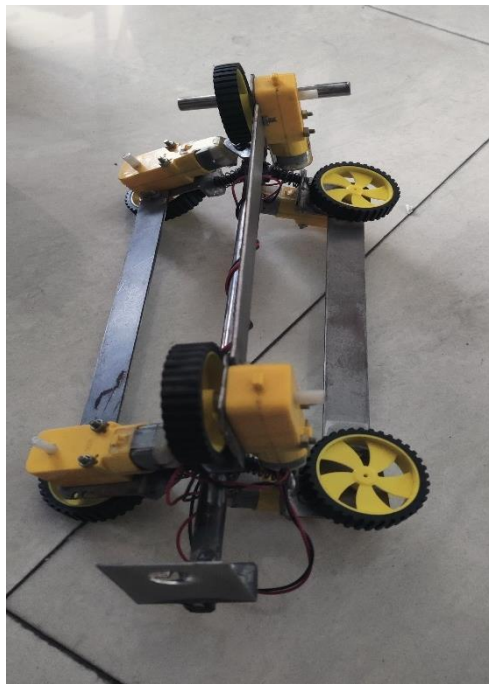


Figure 4.5: Side View of Robot Prototype

Displays the chassis layout with the six-wheel configuration, ensuring even weight distribution and forward movement control.

This mechanical architecture enables the robot to maintain balance and traction inside metallic or concrete pipes while ensuring the central camera has a clear and unobstructed forward view. The use of spring mechanisms allows the wheels to

adapt to minor irregularities inside the pipe surface, improving motion stability during inspection.

2. RASPBERRY PI 5

In this project, the Raspberry Pi 5 serves as the central processing unit for the Smart In-Pipe Inspection Robot. Its 64-bit quad-core Arm Cortex-A76 processor at 2.4GHz and VideoCore VII GPU enable real-time execution of advanced computer vision algorithms such as YOLOv5, critical for detecting corrosion, cracks, and other pipeline defects with high accuracy.



Figure 4.6: Raspberry PI 5

The RP1 southbridge, developed in-house by Raspberry Pi, provides high-speed interfaces necessary for seamless integration of sensors and peripherals. The dual four-lane MIPI transceivers support high-resolution cameras used for visual inspection, while the PCIe 2.0 interface enables expansion to high-speed storage or AI accelerators (e.g., Coral TPU, NVMe SSD).

Key features leveraged in this application include:

- Hardware-accelerated 4Kp60 video decoding for real-time image processing
- LPDDR4X RAM (up to 16GB) for handling deep learning model inference
- USB 3.0 ports for interfacing with external drives or additional sensors
- Gigabit Ethernet for high-speed data transfer during tethered operations or firmware updates

- RTC with battery support to maintain timekeeping in the field
- 5V/5A USB-C PD input, supporting robust power delivery in industrial environments

This makes the Raspberry Pi 5 not only a powerful processing platform but also a compact and cost-effective embedded controller for autonomous pipeline inspection in constrained and hazardous environments.

3. WEB CAMERA

The smart in-pipe inspection robot employs a USB web camera as its primary vision sensor for capturing internal pipeline conditions. The web camera is mounted securely on the robot chassis and aligned in the direction of motion to provide a continuous visual feed of the inner pipe wall. Unlike IP cameras that use Ethernet or Wi-Fi for connectivity, the web camera used here connects directly to the onboard processing unit (such as a Raspberry Pi) via a USB 2.0 or USB 3.0 interface.



Figure 4.6: Web Camera

This real-time video feed is essential for performing on-the-fly defect detection using a trained YOLOv5 object detection model. The USB webcam streams its images directly to the system, allowing the computer to process frames in real time. USB cameras are preferred in embedded systems due to their plug-and-play compatibility with standard computer hardware, compact design, and reliable data transfer rates. This makes them ideal for robotics applications where ease of integration and live imaging are critical.

The selected webcam supports high frame rates and resolution sufficient for detecting fine surface anomalies like cracks and corrosion within pipelines. It operates continuously during robot traversal, enabling real-time visual inspection and logging of detected defects.

4. DC MOTOR

A DC motor is a type of rotary electrical machine that converts direct current (DC) electrical energy into mechanical energy. It operates on the fundamental principle of electromagnetism: when electric current flows through coils of wire inside the motor, it generates a magnetic field that interacts with the static magnetic field of permanent magnets, producing a torque that causes the shaft or armature to rotate. This process effectively transforms electromagnetic energy into rotational motion.



Figure 4.7: DC Motor

In this project, DC motors are utilized to drive the locomotion of the in-pipe inspection robot. These motors provide bidirectional control and variable speed capability, enabling precise and reliable navigation within confined pipeline environments. Their compact form factor makes them well-suited for in-pipe applications where space is limited.

DC motors operate based on electromagnetic principles—when current flows through the internal coil windings, it interacts with the magnetic field from permanent magnets, generating torque that causes the motor shaft to rotate. To sustain continuous rotation, most DC motors incorporate an internal commutation system, which can be either mechanical (using brushes and a commutator) or electronic.

Due to their simplicity, efficiency, and ease of speed control, DC motors have long been a staple in numerous applications including robotics, automation systems, consumer electronics, and mobile devices. Their speed can be precisely adjusted by varying the supply voltage or regulating the current through their windings, which makes them ideal for robotic motion control systems like the one used in this project.

5. RELAY:

A relay is an electrically operated switch used in the in-pipe inspection robot to control high-power devices (like motors or lights) using low-power digital signals from the microcontroller or Raspberry Pi.



Figure 4.8: Relay

In this system, the relay acts as an interface between the control unit and the DC motors or auxiliary systems, allowing isolation between the low-power control logic and the high-power motor circuits. This ensures safe and efficient control over the robot's mechanical components.

Working Principle:

The relay consists of:

- Electromagnet (coil): Energized using a small current from the controller.
- Armature (switching mechanism): Moves to open or close the high-power circuit.
- Contacts (NO/NC): Connects to the external device like the motor.

When the control board sends a signal, the coil is energized, creating a magnetic field that moves the armature. This action switches the connected high-power circuit (e.g., motor on/off).

Specifications:

- Type: Electromechanical relay (5V or 12V depending on design)
- Contacts: SPDT or DPDT (Single/Double Pole Double Throw)
- Rated Load: Typically supports up to 10A at 250V AC or 30V DC
- Control Signal: 3.3V or 5V TTL from Raspberry Pi or microcontroller
- Isolation: Provides electrical isolation between control and load circuits

6. BATTERY (POWER SOURCE):

The in-pipe inspection robot is powered by a rechargeable battery pack consisting of four 18650 Lithium-ion (Li-ion) cells, which collectively serve as the main energy source for all onboard electronic components. These batteries are chosen for their high energy density, lightweight, and reliable discharge characteristics, making them well-suited for compact mobile robotic applications.



Figure 4.9: 18650 Lithium-ion (Li-ion) cells

Specifications:

- Battery Type: 18650 Li-ion cylindrical cells (x4)
- Nominal Voltage per Cell: 3.7V
- Total Voltage:
 - a. Series Configuration (4S): 14.8V (used for high-power applications)
 - b. Parallel Configuration (4P): 3.7V with 4x capacity (for longer runtime)

- c. Series-Parallel (2S2P): 7.4V at double capacity (commonly used)
- Capacity per Cell: 2200–3000mAh
- Total Capacity (2S2P Example): ~4400–6000mAh
- Discharge Rate: Suitable for moderate loads such as Raspberry Pi and DC motors
- Charging Method: Balanced charging using Li-ion battery charger (e.g., TP4056 modules or BMS-based chargers)

4.3. Software Architecture

The software architecture of the In-Pipe Inspection Robot is designed to integrate real-time hardware control, image acquisition, defect detection using a deep learning model, and user interaction via a web interface. It is modular, scalable, and optimized for performance on a resource-constrained embedded device (Raspberry Pi 5).

1. YOLO v5

YOLOv5 (You Only Look Once version 5) is an advanced, real-time object detection algorithm developed in PyTorch. It improves upon its predecessors (YOLOv3/YOLOv4) with faster training, higher accuracy, modular design, and better integration with modern deep learning pipelines. In this project, YOLOv5 is used to detect two classes of pipeline defects: crack and corrosion.

Key Concepts:

YOLOv5 treats object detection as a single regression problem. Instead of using separate networks for object proposal, classification, and refinement (like Faster R-CNN), YOLOv5 predicts bounding boxes and class labels in a single pass, making it extremely fast.

YOLOv5 Architecture Components:

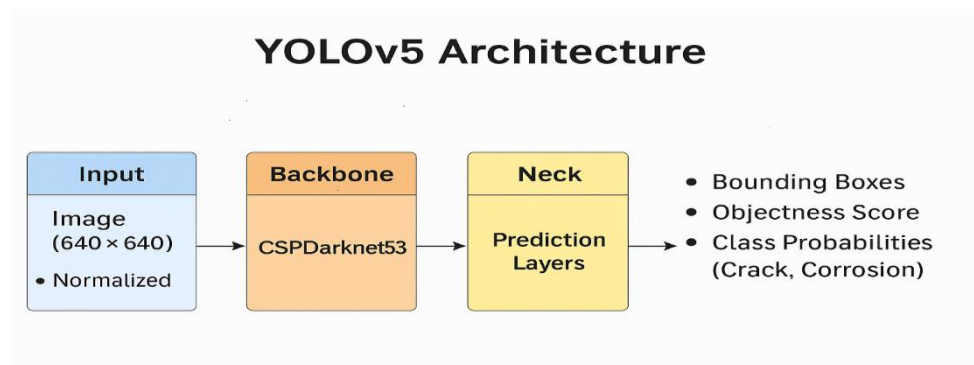


Figure 4.10: YOLOv5 Architecture

1. Input:

- Resized image (typically $640 \times 640 \times 3$)
- Normalized to $[0, 1]$ pixel range
- Format: $[\text{batch_size}, \text{channels}, \text{height}, \text{width}]$

2. Backbone – CSPDarknet53

- Extracts spatial and semantic features from the image
- Uses Cross Stage Partial (CSP) connections to improve learning efficiency and reduce computation

3. Neck – PANet + FPN

- Enhances feature fusion from different scales (low $\rightarrow$ high resolution)
- Helps detect small, medium, and large objects in the same image

4. Head – Dense detection layers

- Predicts bounding boxes, objectness scores, and class probabilities at three scales (e.g., 80×80 , 40×40 , 20×20 grids)

Prediction Output:

At each grid cell and anchor box, YOLOv5 predicts:

- 4 bounding box parameters (center x, center y, width, height)
- 1 objectness score (probability that an object exists in the box)

C class scores (softmax/sigmoid over all defined classes, e.g., crack, corrosion)

Total prediction per anchor box: $(4 + 1 + C)$ values

YOLOv5 uses anchor-based detection, meaning it predicts offsets relative to predefined box shapes (anchors) at each grid scale.

YOLOv5 Inference Algorithm:

Input: RGB image resized to 640×640

Output: List of detected objects with class, confidence, and bounding box

Step 1: Preprocessing

- Resize input image to 640×640 pixels
- Normalize pixel values to $[0, 1]$
- Convert from BGR to RGB
- Rearrange shape to $[\text{batch_size}, \text{channels}, \text{height}, \text{width}]$

Step 2: Forward Pass Through YOLOv5 Model

- Pass image tensor through the backbone (e.g., CSPDarknet53) for feature extraction
- Pass features through PANet + FPN neck for multi-scale fusion
- Output is a detection matrix from head (predictions at 3 scales)

Step 3: Prediction Decoding

- For each anchor box at each scale:
- Decode bounding box coordinates
- Compute objectness score $\times$ class confidence
- Apply sigmoid activation for classification and objectness outputs

Step 4: Non-Maximum Suppression (NMS)

- For overlapping boxes with the same class, retain the one with the highest confidence

- Remove redundant boxes (based on IoU threshold, e.g., 0.45)

Step 5: Postprocessing

- Rescale bounding boxes to original image dimensions
- Filter out detections below confidence threshold (e.g., 0.5)

Key Features of YOLOv5:

- Real-Time Speed: Suitable for Raspberry Pi (~4–5 FPS)
- Anchor-based Prediction: Predefined box shapes help detect objects of various sizes
- Multi-scale Detection: Identifies small cracks and large corrosion areas simultaneously
- Lightweight Model (YOLOv5s): ~7M parameters, ~13MB TorchScript export size

In this project, YOLOv5s was used due to its balance of speed, accuracy, and computational efficiency, making it ideal for embedded inspection applications in constrained environments.

2. Open CV

OpenCV is a powerful open-source computer vision and image processing library used in this project for real-time camera input handling and pre-processing before feeding frames into the YOLOv5 model.

Key Functions of OpenCV:

- a. Real-Time Image Acquisition
 - i. Captures video frames from the Pi Camera using `cv2.VideoCapture()`.
 - ii. Converts frame formats (e.g., BGR to RGB) compatible with YOLOv5 input.
- b. Image Pre-Processing

- i. Resizing: Frames are resized to 640×640 to match YOLOv5 input requirements.
 - ii. Normalization: Pixel values are normalized to a 0–1 range.
 - iii. Color Space Conversion: Converts raw BGR frames to RGB format for inference.
- c. Visualization
 - i. Draws bounding boxes, labels, and confidence scores on detected defects using `cv2.rectangle()` and `cv2.putText()`.
 - ii. Displays the final output in both the Flask interface and local preview window (for debugging).
- d. Performance Optimization
 - i. Reduces frame resolution for faster processing if needed.
 - ii. Can apply filtering (blurring or sharpening) to enhance detection under noisy lighting conditions.

OpenCV is chosen for this project due to its efficiency and suitability for embedded systems like the Raspberry Pi. It offers real-time image processing capabilities with minimal latency, which is essential for timely defect detection in dynamic inspection environments. Its lightweight nature ensures smooth performance even on resource-constrained hardware. Furthermore, OpenCV seamlessly integrates with deep learning models such as YOLOv5, enabling streamlined workflows for object detection and visualization. It supports a wide range of functionalities, from capturing video frames to rendering graphical outputs, making it ideal for both backend processing and front-end GUI display in robotic inspection applications.

3. Python

Python is the primary programming language used in this project due to its simplicity, readability, and extensive support for robotics, machine learning, and hardware interfacing. It enables seamless integration of different modules, including motor control via GPIO pins, image processing with OpenCV, machine learning inference using the YOLOv5 model in PyTorch, and web interface

development through the Flask framework. Python's rich ecosystem of libraries and tools allows rapid development and testing of both software and hardware components. Its cross-platform compatibility and strong community support make it particularly well-suited for embedded systems like the Raspberry Pi. In this project, Python acts as the central glue connecting hardware control, real-time computer vision, and user interface functionality into a cohesive and efficient robotic inspection system.

4. Flask (Python Web Framework)

Flask is a lightweight Python web framework used in this project to build a real-time monitoring interface for the In-Pipe Inspection Robot. It acts as the backend server running on the Raspberry Pi, enabling the system to host a local web application that streams video frames processed by the YOLOv5 model. Flask handles routing, template rendering, and communication between the robot's detection logic and the user-facing dashboard. The processed frames with bounding boxes are rendered and displayed in the browser, allowing users to view defect detections in real time without needing any additional software. Flask's simplicity, modularity, and compatibility with embedded platforms make it an ideal choice for deploying interactive web applications on low-power devices. Its integration in this project provides a user-friendly interface for monitoring pipeline inspections remotely and efficiently.

5. HTML, CSS, and JavaScript

HTML, CSS, and JavaScript are used to design and develop the front-end interface of the web dashboard hosted by the Flask server. HTML provides the structure of the web pages, defining elements such as video display windows, headers, and status indicators. CSS is used to style these elements, ensuring the interface is clean, responsive, and visually intuitive for users monitoring pipeline inspections. JavaScript enhances the interactivity of the dashboard by enabling features such as live video streaming, automatic refresh of detection results, and real-time feedback updates. This combination of front-end technologies ensures that users can easily view and interpret defect detection outputs from the robot in a browser, even on mobile or low-resource devices. Together, HTML, CSS, and JavaScript contribute

to a responsive and accessible user interface for remote and real-time inspection monitoring.

6. Raspberry Pi OS

Raspberry Pi OS is a lightweight, Debian-based operating system specifically optimized for Raspberry Pi hardware. In this project, it serves as the primary operating system for the Raspberry Pi 5, providing a stable and efficient environment for running all components of the in-pipe inspection system, including Python scripts, the YOLOv5 object detection model, and the Flask web server. Raspberry Pi OS supports essential libraries such as OpenCV, PyTorch, and GPIO control modules, making it highly suitable for robotics and machine learning applications. Its compatibility with various development tools and ease of integration with both hardware and software components make it ideal for deploying embedded AI solutions. The OS also offers a user-friendly interface and remote access capabilities, enabling developers to monitor and debug the system during testing and deployment phases.

7. GPIO Library (RPi.GPIO)

The RPi.GPIO library is a Python module that provides an easy-to-use interface for controlling the General-Purpose Input/Output (GPIO) pins on the Raspberry Pi. In this project, it is used to control the DC motors through a relay module enabling basic robotic movement such as forward motion and stopping. relays are used to switch motor circuits ON or OFF, allowing the Raspberry Pi to control motor movement (e.g., forward, stop) through GPIO signals. The RPi.GPIO library allows setting up GPIO pins as outputs and toggling their state (HIGH/LOW) to energize or de-energize the relay coils. Its simplicity and compatibility with Raspberry Pi OS make it ideal for real-time hardware interaction. By using RPi.GPIO, the robot can execute precise movement commands in synchronization with the defect detection system, ensuring coordinated operation during in-pipe inspections.

CHAPTER 5

SYSTEM DESIGN

This chapter describes how the smart in pipe inspection robot is designed to work as a complete system. It explains the structure of both hardware and software parts and how they are connected to perform tasks like movement, video capture, and defect detection. The overall system flow and functional blocks are also explained in detail

5.1. System Interaction

The system interaction diagram provides a high-level overview of the sequence of operations and interactions among various components of the In-Pipe Inspection Robot, from initialization to real-time monitoring and inspection reporting. It outlines how the operator, hardware modules (Raspberry Pi, Camera, Motor Driver, DC Motors), AI model (YOLOv5), and logging system work in coordination throughout the inspection lifecycle.

The operation of the smart in-pipe inspection robot begins with system initialization, where the operator powers on the robot and the Raspberry Pi sets up the camera interface, motor control modules, and loads the YOLOv5 model into memory for real-time inference. Once initialized, the operator starts the inspection process, prompting the Raspberry Pi to activate the DC motors through the motor driver, allowing the robot to move forward inside the pipeline. During traversal, the robot enters a continuous monitoring phase in which the camera captures video frames that are processed in real time by the YOLOv5 model. The model identifies defects such as cracks or corrosion and provides confidence scores. If a defect is detected, the system estimates its location based on time or distance traveled, logs the defect type, confidence, and position. If no defect is found, the corresponding pipe segment is logged as defect-free. Finally, the inspection process ends either manually by the operator or automatically after completing the assigned distance. The motors are stopped, data is finalized, and a comprehensive inspection report is generated for review.

The smart in-pipe inspection robot is composed of several key components, each playing a specific role in the overall system operation. The operator is responsible for initiating and supervising the inspection process. At the core of the system is the Raspberry Pi, which acts as the central controller, managing hardware operations, executing the defect detection logic, and handling the web-based user interface. The camera module is mounted on the robot to continuously capture visuals from within the pipeline, providing the input for real-

time analysis. These captured frames are processed by the YOLOv5 model, which detects cracks and corrosion with high accuracy and speed. For movement, the robot relies on a motor driver and DC motors, which enable forward motion and controlled navigation inside the pipeline. Finally, a data logger component stores or transmits information about detected defects, including their type, confidence level, and approximate location, ensuring that a complete inspection record is available for later reporting and review.

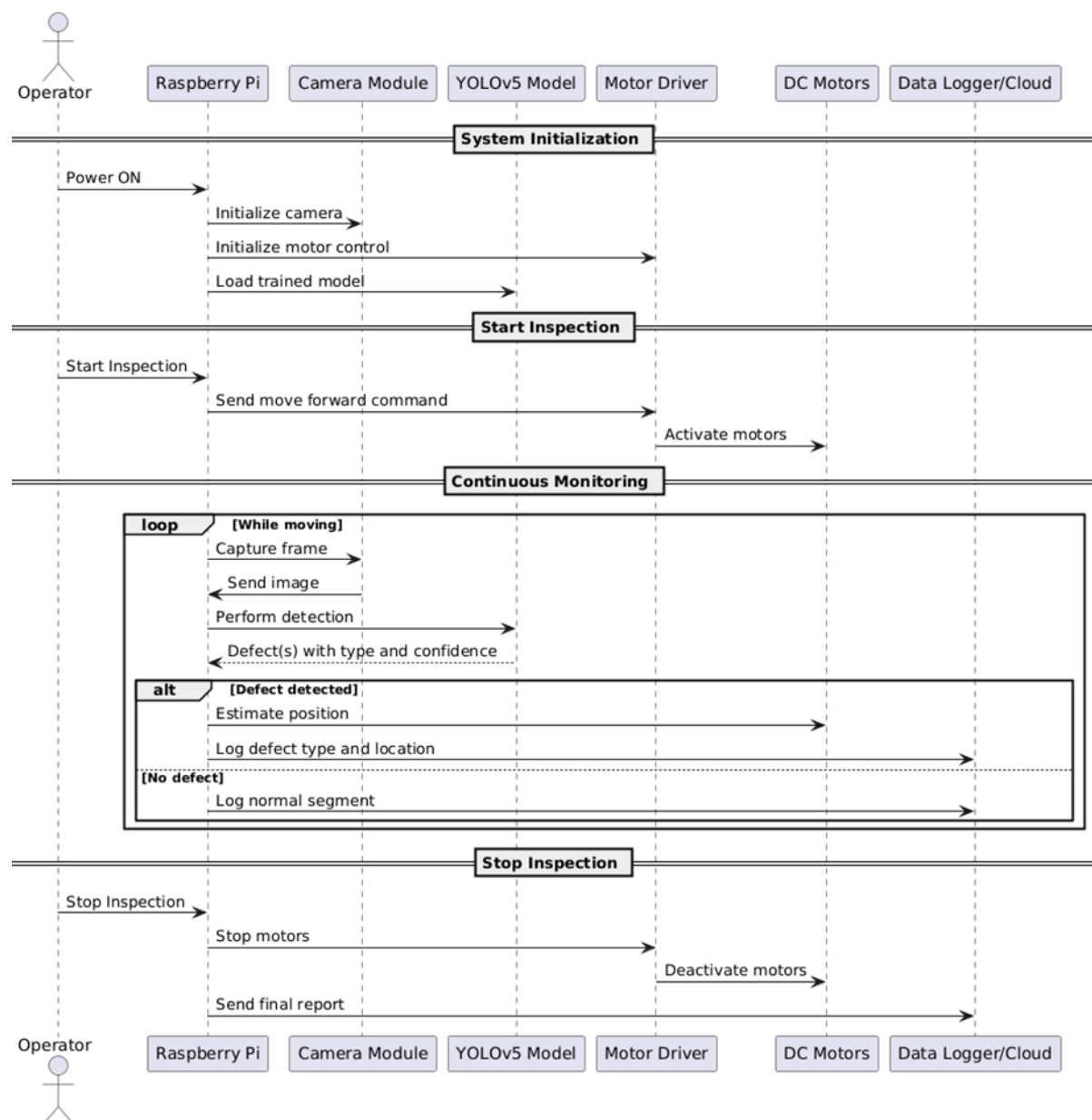


Figure 5.1: System Interaction Diagram

5.2. Use Case Diagram

The Use Case Diagram illustrates the interaction between the Operator and the Pipe Inspection Robot System. It outlines the key functionalities that the operator engages with and how the internal subsystems function during the pipeline inspection process. This diagram reflects a local-only system (no cloud reporting), where results are stored and displayed locally via a web dashboard.

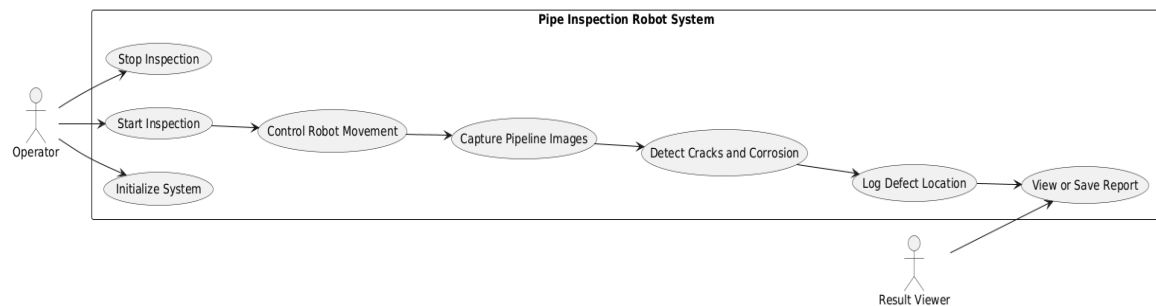


Figure 5.2: Use Case Diagram

The smart in-pipe inspection system involves two main actors: the operator and the result viewer. The operator is responsible for starting, monitoring, and stopping the inspection process, while the result viewer—who could be the operator or a separate technician—reviews the inspection results and reports.

The system supports several core use cases. It begins with the operator initializing the system by powering on the Raspberry Pi, which sets up the camera interface, motor controls, and loads the YOLOv5 detection model into memory. Once initialized, the operator starts the inspection, triggering the robot's real-time operation and enabling both movement and data collection. During the inspection, the Raspberry Pi sends commands to the motor driver to control the robot's movement through the pipe, which can be operated manually or automated based on the requirement. As the robot navigates, the Pi Camera continuously captures video frames of the internal pipeline surface. These frames are then processed by the YOLOv5 model to detect and classify defects such as cracks or corrosion. If a defect is identified, the system logs the event along with an estimated location based on the time and distance of traversal. The inspection process ends when the operator stops the robot. Finally, a report summarizing the detected defects is generated, stored locally, and made accessible through a Flask-based web dashboard for review and analysis.

5.3. Data Flow Diagram

The Data Flow Diagram shown above illustrates the internal and external data interactions within the In-Pipe Inspection Robot System. The system is built around a Raspberry Pi, which integrates computer vision, motor control, and defect logging modules. This diagram explains how data flows between these modules, hardware components, and the operator during an inspection run.

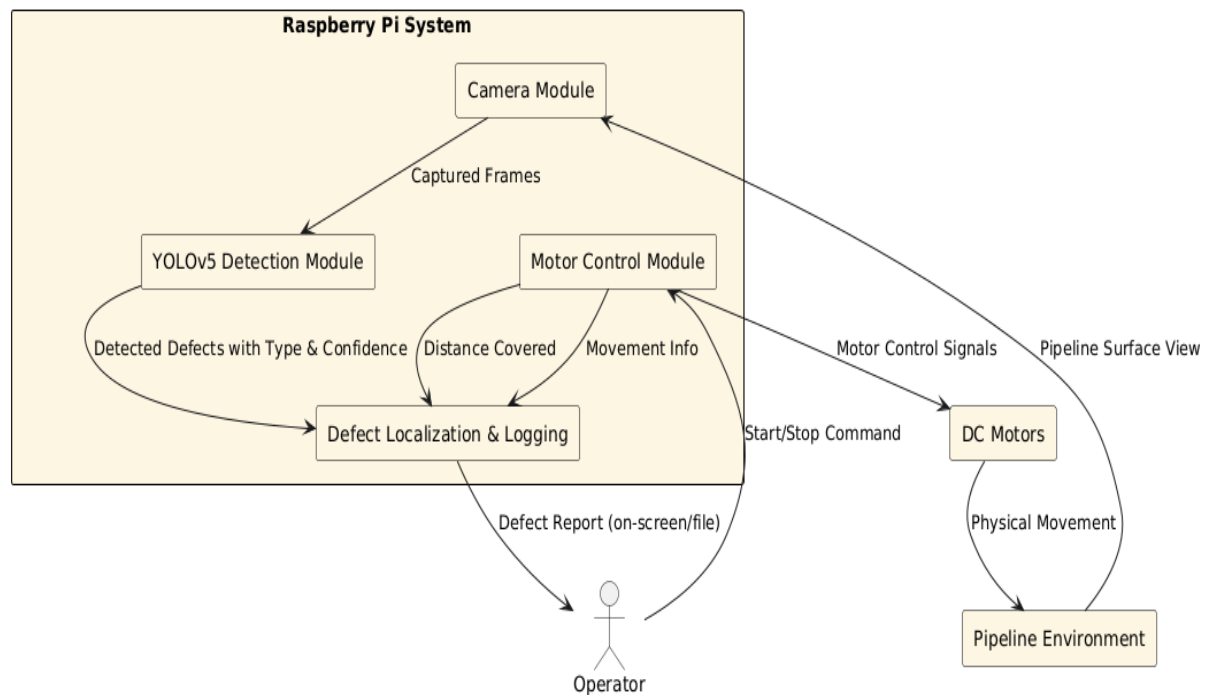


Figure 5.3: Data Flow Diagram

System Overview: At the core of the system is the Raspberry Pi, which manages the robot's entire operation. It communicates with the camera, motor drivers, and onboard AI model to detect pipeline defects and logs results for operator review.

Component Interactions: The smart in-pipe inspection robot consists of multiple components that interact seamlessly to perform real-time defect detection and logging. The process begins when the operator sends a start or stop command to the Raspberry Pi system, which acts as the central controller. As the robot begins to move, the camera module continuously captures live video frames of the pipeline's inner surface. These frames are immediately forwarded to the YOLOv5 detection module, which analyzes each frame to identify defects such as cracks or corrosion. The model outputs the defect type along with

a confidence score, and this information is passed to the defect localization and logging module.

Simultaneously, the motor control module receives navigation commands either directly from the operator or from the robot's automated logic. It translates these commands into electrical signals that are sent to the DC motors, which in turn generate the physical movement of the robot through the pipeline. As the robot moves, its changing position results in dynamic updates to the camera's field of view, which influences the continuous flow of visual data being analyzed. The motor control module also tracks and logs the distance covered during operation, enabling accurate mapping of defects relative to the robot's position.

The defect localization and logging module then correlates the identified defect information with the robot's movement data to estimate the precise location of each detected issue within the pipeline. Finally, a comprehensive defect report is generated, which can either be displayed on a screen in real time or saved as a file for further review and analysis.

5.4. Activity Diagram

The Activity Diagram provides a step-by-step visual representation of the operational workflow of the In-Pipe Inspection Robot. It outlines the logical sequence of activities performed from system startup to shut down, focusing on the coordination between motion control, image processing, defect detection, and logging.

The process begins when the Raspberry Pi is powered on, and all required subsystems are initialized. This includes setting up the Pi Camera, loading the trained YOLOv5 model, and configuring GPIO pins to control the DC motors via the motor driver.

Once initialized, the system waits for a start command from the operator, which is typically issued through a local web interface. Upon receiving this command, the robot starts moving forward inside the pipeline, and a continuous loop of inspection begins.

During each cycle of the inspection loop, the Pi Camera captures a frame of the pipeline interior as the robot moves forward. This frame is then pre-processed using OpenCV, which involves operations such as resizing, normalization, and color space conversion to ensure compatibility with the detection model. The processed image is passed to the YOLOv5 model for inference. The model analyzes the frame and determines whether any defects, such as cracks or corrosion, are present. If a defect is detected, it is logged along with its

type, confidence score, and estimated position based on the robot's movement. If no defect is found, the system optionally logs the segment as clear, which helps in maintaining a complete inspection record of the entire pipeline.

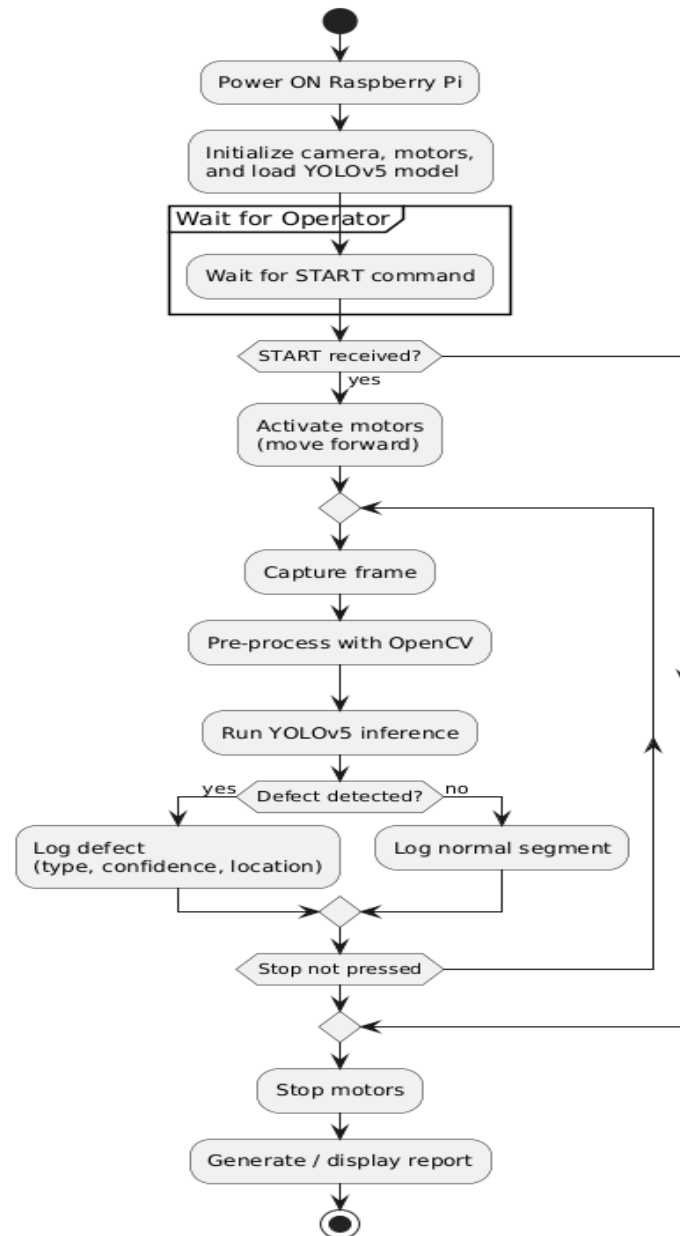


Figure 5.4: Activity Diagram

This inspection loop continues until the operator issues a stop command or the robot completes its planned path. At this point, the system stops the motors and proceeds to generate a final report, summarizing all defect detections and their corresponding metadata. The report is stored locally and may be accessed through the same web interface.

Figure 5.4 highlights the system’s automation, real-time feedback, and minimal reliance on human intervention, making it suitable for autonomous inspection of hazardous or inaccessible pipeline environments.

5.5. Class Diagram

The Class Diagram illustrates the modular software structure of the In-Pipe Inspection Robot. It emphasizes object-oriented design principles, where each class is dedicated to a specific functional responsibility, ensuring separation of concerns, reusability, and easier debugging.

At the center of the system is the RobotController class, which coordinates the interaction between all subsystems, including the camera, motors, machine learning model, logging system, and the web interface.

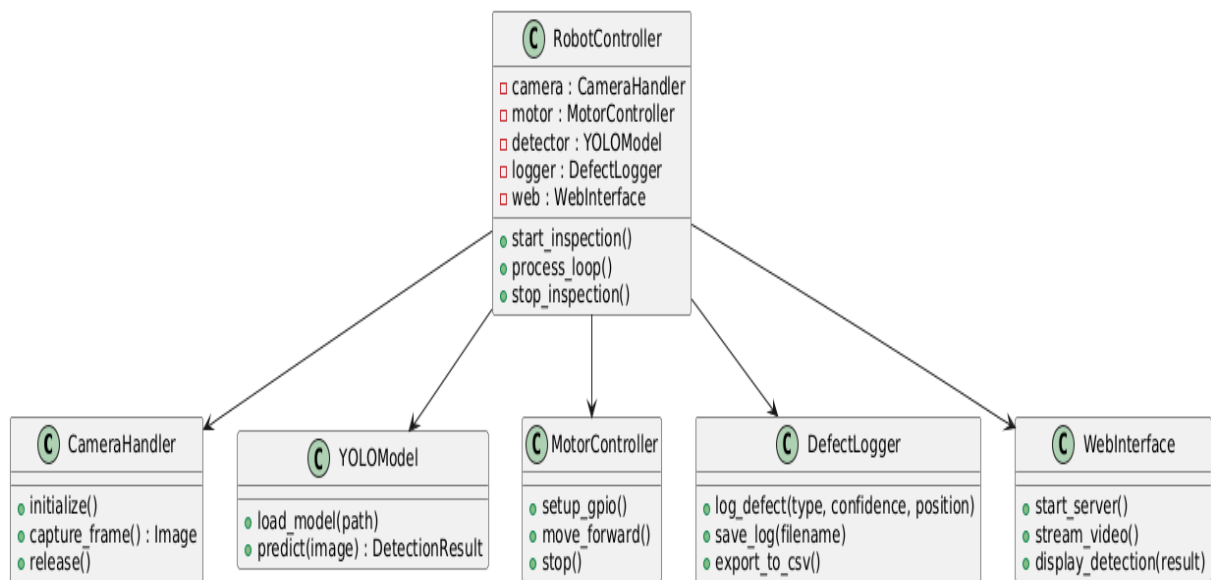


Figure 5.5: Class Diagram

The software architecture of the smart in-pipe inspection robot is organized into modular classes, each responsible for a specific function. At the center of the system is the **RobotController** class, which acts as the main control unit. It manages and initializes all other key classes including **CameraHandler**, **YOLOModel**, **MotorController**, **DefectLogger**, and **WebInterface**. The **RobotController** contains core methods such as `start_inspection()`, which initiates the inspection process; `process_loop()`, which runs the main loop by capturing frames, detecting defects, and logging results; and `stop_inspection()`, which safely terminates the session and finalizes outputs.

The CameraHandler class interfaces directly with the Pi Camera module. It provides methods like initialize() to set up the camera, capture_frame() to take real-time images for analysis, and release() to safely close the camera interface after the inspection ends. The defect detection task is handled by the YOLOModel class, which loads and executes the YOLOv5 deep learning model. It includes methods such as load_model(path) for loading a pretrained model and predict(image) for performing inference on captured frames, returning bounding boxes, defect types, and confidence scores.

Motor control is managed by the MotorController class, which uses the Raspberry Pi's GPIO pins to control the robot's movement. It includes methods such as setup_gpio() for configuring motor pins, move_forward() for driving the robot forward, and stop() to halt the robot. The DefectLogger class is responsible for organizing and storing inspection data. It provides functionality through methods like log_defect(type, confidence, position) to record each detected issue, save_log(filename) to save the results to a structured file format like CSV or JSON, and export_to_csv() for exporting the data for further review or analysis.

Finally, the WebInterface class enables real-time interaction through a Flask-based web dashboard. It includes methods like start_server() to launch the web application, stream_video() to display a live video feed with detection overlays, and display_detection(result) to show the detection output to the operator during inspection.

This figure 5.5 showcases a high-cohesion, low-coupling architecture that supports future scalability (e.g., cloud reporting, autonomous navigation) and maintains a clean separation between image capture, AI inference, hardware control, data logging, and user interaction. The modular design also allows unit testing of individual classes for robustness and reliability.

CHAPTER 6

IMPLEMENTATION

This chapter presents the detailed implementation process of the Smart In-Pipe Inspection Robot, including dataset creation, model training, software development, hardware integration, and real-time inference pipeline on Raspberry Pi 5.

6.1. Development Environment

The development of the Smart In-Pipe Inspection Robot was carried out using both cloud-based tools for machine learning model training and embedded platforms for real-time execution. The hardware and software environments used are listed below:

Table 6.1: Development Environment

Component	Specification / Tool
Model Training	Google Colab (with free GPU, PyTorch, and YOLOv5)
Dataset Platform	Roboflow (custom dataset creation and annotation)
Edge Device	Raspberry Pi 5 (4 GB RAM, Quad-core Cortex-A76)
Operating System	Raspberry Pi OS 64-bit (Bookworm)
Programming Language	Python 3.11
ML Framework	YOLOv5 (v6.2)
Libraries Used	OpenCV 4.9, Flask 3.0, RPi.GPIO, Torch, NumPy
Camera Module	USB Web Camera (forward-facing, pipe-mounted)
Motor Driver	Relay
Remote GUI	RealVNC Viewer

6.2. Hardware Integration

The hardware setup of the in-pipe inspection robot was designed to be compact, robust, and suitable for navigating narrow pipe environments while capturing visual data for crack and corrosion detection.

1. COMPONENTS USED:

The robot integrates several compact and efficient components tailored for in-pipe inspection tasks. At the core of the system is the Raspberry Pi, which functions as the main controller responsible for processing image streams and executing the YOLOv5 model for real-time defect detection. A USB web camera, equipped with infrared (IR) LEDs, is used to capture clear visuals inside the pipe, even under low-light conditions. For movement, the robot is driven by three DC geared motors mounted on the front wheels, enabling it to traverse through the pipeline. Three rear wheels is included as a free-rolling dummy wheels, which is not powered, due to power supply constraints. Motor direction and speed are controlled by a relay module that receives control signals via the Raspberry Pi's GPIO pins. The entire system is powered by a battery pack consisting of four 18650 lithium-ion cells, which provide portable and sufficient power to both the Raspberry Pi and motor control circuitry.

2. MECHANICAL DESIGN:

The mechanical structure of the robot is built on a metallic chassis, onto which a wooden platform is securely mounted to hold all major components. This platform accommodates the Raspberry Pi, motor control board, relay module, and battery pack in a compact and organized layout. The robot's locomotion is driven by three front wheels, each connected to a geared DC motor that provides both forward motion and turning capability within the pipeline. At the rear, a set of three undriven wheels act as passive support, helping to balance the robot while minimizing power consumption and load on the motors. To ensure clear visual coverage of the pipeline's interior, the camera is fixed at the front of the robot using a metal extension bracket, allowing for stable and direct imaging during inspection.



Figure 6.1: Front View of the Robot (Camera Mounted)



Figure 6.2: Camera Unit with IR LEDs for Low-Light Visibility

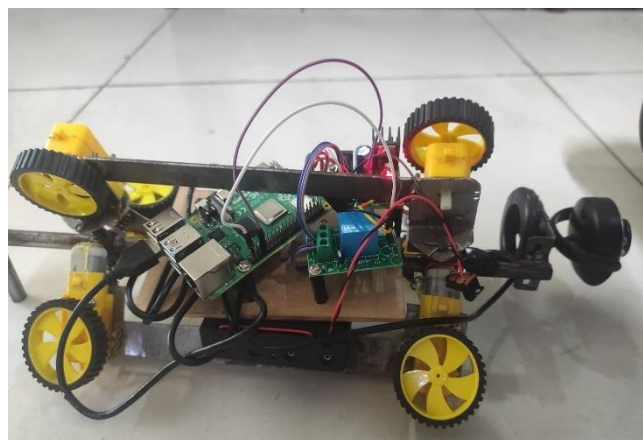


Figure 6.3 Side View Showing Motor Wiring and Chassis

3. ELECTRICAL INTEGRATION:

The GPIO pins of the Raspberry Pi are used to control the relay module or motor driver, allowing the system to operate based on user commands or autonomous logic during inspection. A regulated battery pack is employed to provide a shared power supply to both the Raspberry Pi and the motors, ensuring stable operation throughout the inspection process. All internal wiring is carefully arranged to minimize electrical interference and signal noise. Connections between components are established using a combination of jumper wires, terminal blocks, and soldered joints, offering both flexibility and reliability during movement and data processing inside the pipeline.

Due to constraints in available power, only the front three wheels are powered while the rear remains passive. Despite this, the robot is capable of stable in-pipe movement and capturing inspection footage.

6.3. Dataset Preparation

The dataset titled "In-Pipe Corrosion and Cracks" was completely created and manually annotated using Roboflow. It includes a total of 1,073 labeled images, divided as:

Subset Images Percentages

Table 6.2: Dataset Split

Dataset Split	Number of Images	Percentage
Train Set	879	82%
Valid Set	122	11%
Test Set	72	7%

The custom dataset used for training the defect detection model consisted of two defined classes: class 0 for cracks and class 1 for corrosion. Before training, all images underwent a preprocessing stage to standardize and enhance input consistency. Auto-orientation was enabled to ensure that each image was aligned correctly, regardless of how it was captured. Each image was then resized and stretched to a fixed resolution of 640×640 pixels, as required by the YOLOv5 input layer.

To improve model robustness and prevent overfitting, a series of data augmentation techniques were applied during training. These included vertical flipping and 90-degree rotations in both clockwise and counter-clockwise directions to simulate different camera orientations. A shear transformation of up to ± 15 degrees was introduced both horizontally and vertically to mimic real-world distortions. Additionally, 15% of the dataset images were converted to grayscale to help the model generalize in varying lighting conditions. Finally, a blur filter of up to 2.5 pixels was applied to simulate motion blur or minor defocus, improving detection accuracy under imperfect capture scenarios. All annotations were exported in YOLOv5 format (normalized coordinates in .txt files).

6.4. YOLOv5 Model Training (in Google Colab)

To train the defect detection model, the YOLOv5 framework was used due to its high accuracy, real-time performance, and compatibility with embedded platforms like Raspberry Pi. The training was conducted using Google Colab, which provides access to free GPU resources and seamless integration with Roboflow-exported datasets.

Environment Setup

The training pipeline was implemented in Google Colab, with the following configurations:

- Hardware accelerator: NVIDIA Tesla T4 GPU (provided by Colab)
- Framework: PyTorch 1.13
- YOLO version: YOLOv5 v6.2 (Ultralytics GitHub repository)
- Python version: 3.9
- Dataset format: YOLOv5 format (exported from Roboflow)

Required dependencies were installed via:

```
!git clone https://github.com/ultralytics/yolov5
```

```
%cd yolov5
```

```
!pip install -r requirements.txt
```

Dataset Import (via Roboflow)

The dataset was imported directly from the Roboflow workspace using the Roboflow Python SDK:

```
from roboflow import Roboflow

rf = Roboflow(api_key="YOUR_API_KEY")

project = rf.workspace("in-pipe-corrosion").project("in-pipe-corrosion-and-cracks")

version = project.version(4)

dataset = version.download("yolov5")
```

This downloaded and unpacked the dataset with labeled train, valid, and test folders along with a data.yaml file.

Training Parameters

The training was initiated using the following command:

```
!python train.py --img 640 --batch 16 --epochs 150 --data data.yaml --weights yolov5m.pt
--project project /content/yolov5/runs/train --name custom_training
```

Table 6.3: Training Parameters

Parameter	Value
Model used	YOLOv5m (medium variant)
Input size	640×640
Batch size	16
Epochs	150
Optimizer	SGD
Loss Function	BCE + GIoU

Training YOLOv5 in Google Colab provided a powerful yet cost-effective environment to experiment and iterate on the detection model using GPU acceleration, while Roboflow made dataset integration seamless.

6.5. Software Setup

The software setup is the backbone of the robot's intelligence and functionality. It comprises various layers including camera streaming, GPIO motor control, defect detection using a deep learning model (YOLOv5), and a Flask-based user interface for interaction.

1. Operating System and Development Environment

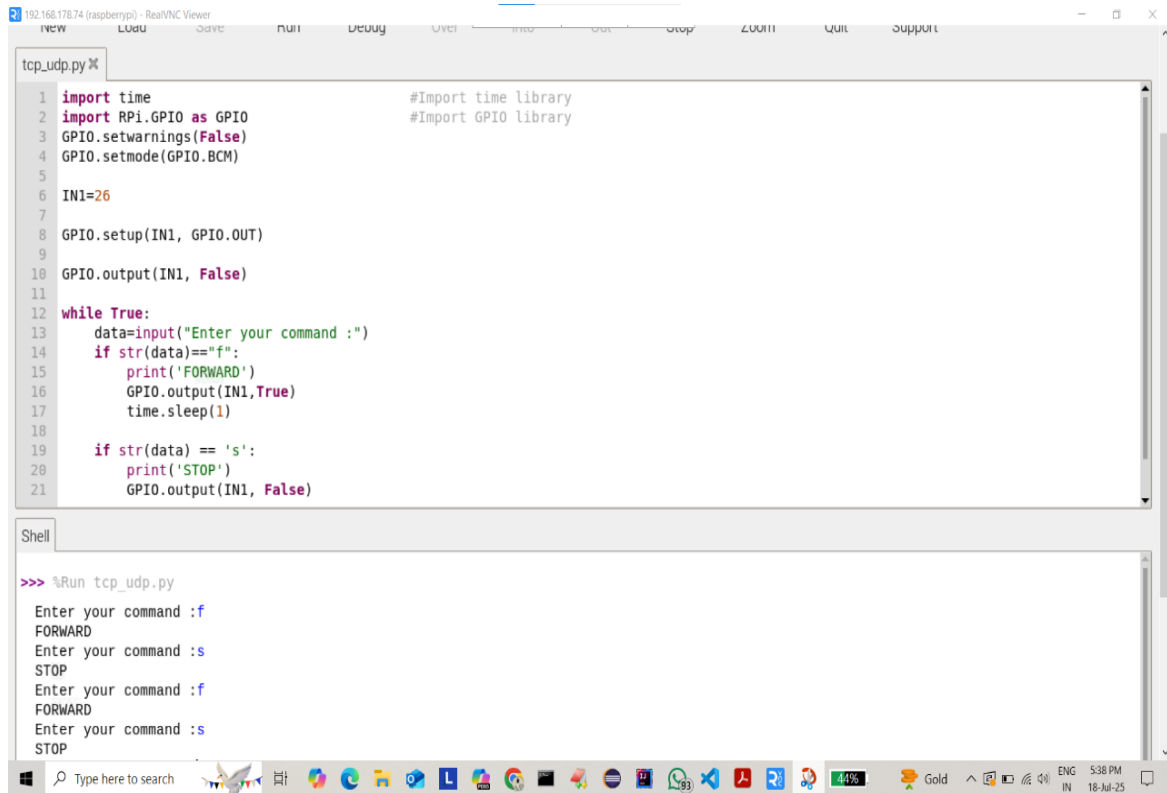
The software environment for the smart in-pipe inspection robot was built on Raspberry Pi OS, running on a Raspberry Pi board. Python 3 was used as the primary programming language due to its flexibility and compatibility with various libraries and hardware interfaces. For development and testing, Thonny IDE was used to implement and debug GPIO-based motor control and real-time logic, while Visual Studio Code (VS Code) was utilized for model development and web application programming. RealVNC was employed to provide GUI-based remote access to the Raspberry Pi, enabling easier development and monitoring.

Several key Python libraries were integrated into the project. The RPi.GPIO library was used to control motor directions through the GPIO pins. OpenCV handled image capture, display, and preprocessing tasks. For defect detection, the Ultralytics YOLOv5 framework was used, supported by essential libraries such as Torch, NumPy, PIL (Python Imaging Library), and Matplotlib for model training, inference, and visualization. Flask was used to create a lightweight web application that allowed the operator to monitor live video feeds and view detection results through a browser-based interface.

2. GPIO Motor Control

A Python script (tcp_udp.py) was developed to test basic robot motion using keyboard commands:

```
if data == "f":  
  
    GPIO.output(IN1, True) # Move Forward  
  
if data == "s":  
  
    GPIO.output(IN1, False) # Stop
```

The image shows a RealVNC Viewer window with a title bar indicating the connection to 192.168.178.74 (raspberrypi). The main window is divided into two panes. The top pane, titled 'tcp_udp.py', contains a Python script for controlling a motor via GPIO. The script imports the 'time' and 'RPI.GPIO' libraries, sets warnings to False and mode to BCM, defines pin IN1 as 26, sets it as an output, and enters a loop where it listens for 'f' (forward) or 's' (stop) commands. The bottom pane, titled 'Shell', shows the execution of the script and the user's interactions: entering 'f' results in 'FORWARD' and entering 's' results in 'STOP'. The Windows taskbar at the bottom shows the time as 5:38 PM on 18-Jul-25.

```
1 import time                                #Import time library
2 import RPi.GPIO as GPIO                    #Import GPIO library
3 GPIO.setwarnings(False)
4 GPIO.setmode(GPIO.BCM)
5
6 IN1=26
7
8 GPIO.setup(IN1, GPIO.OUT)
9
10 GPIO.output(IN1, False)
11
12 while True:
13     data=input("Enter your command :")
14     if str(data)=="f":
15         print('FORWARD')
16         GPIO.output(IN1,True)
17         time.sleep(1)
18
19     if str(data) == 's':
20         print('STOP')
21         GPIO.output(IN1, False)
```

```
>>> %Run tcp_udp.py
Enter your command :f
FORWARD
Enter your command :s
STOP
Enter your command :f
FORWARD
Enter your command :s
STOP
```

Figure 6.4: Motor Control Script and Terminal Output

This image shows the Python script used to control the DC motors through GPIO pins on the Raspberry Pi. The below shell displays the terminal where the user inputs commands such as "f" (forward) and "s" (stop), and the system responds by activating or deactivating the motor relays accordingly.

3. Camera Streaming and Model Integration

The robot uses a USB web camera, mounted at the front, to continuously capture high-resolution images of the pipeline interior. The video stream is accessed on the Raspberry Pi using OpenCV's `cv2.VideoCapture()`, configured at a resolution of 640×480 pixels. For remote monitoring, RealVNC is used to access the Raspberry Pi desktop from another device.

The captured frames are fed into a custom-trained YOLOv5s model to detect defects such as cracks and corrosion. The model is loaded using PyTorch, and real-time inference is performed via a Python script (`livedetect.py`). Frames are resized and normalized before detection, and results are displayed with bounding boxes, class labels, and confidence scores using OpenCV.

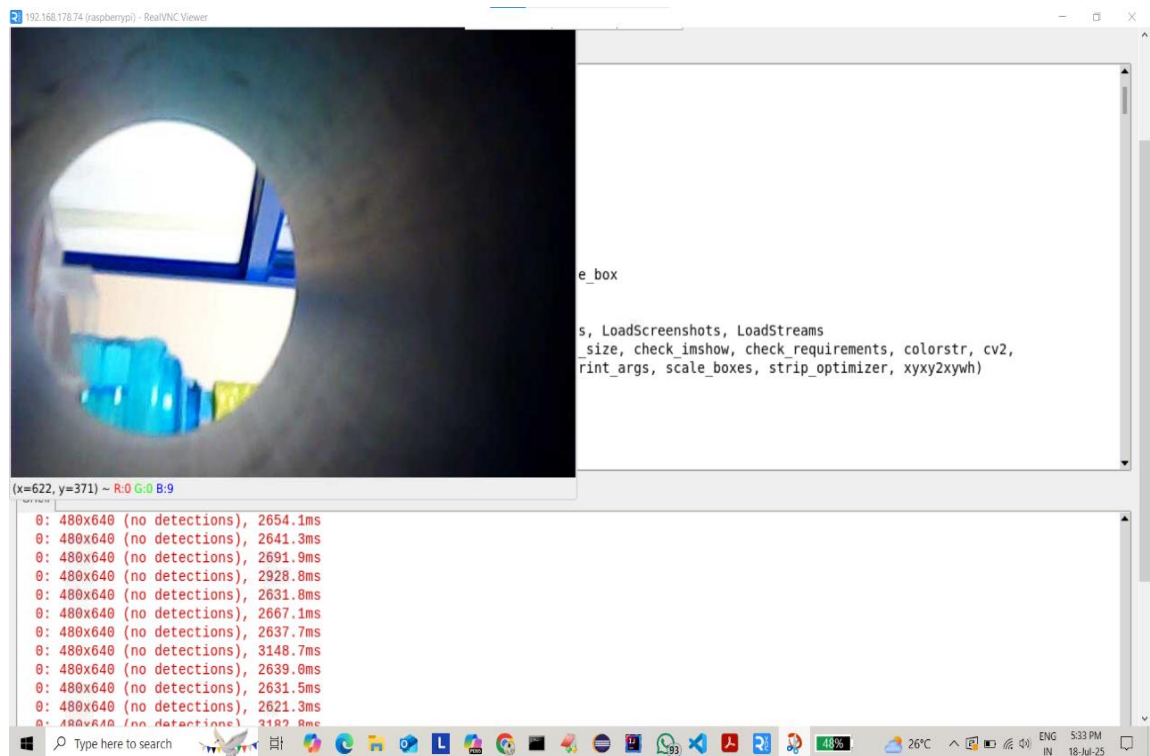


Figure 6.5: Live camera feed viewed through VNC showing internal pipe surface

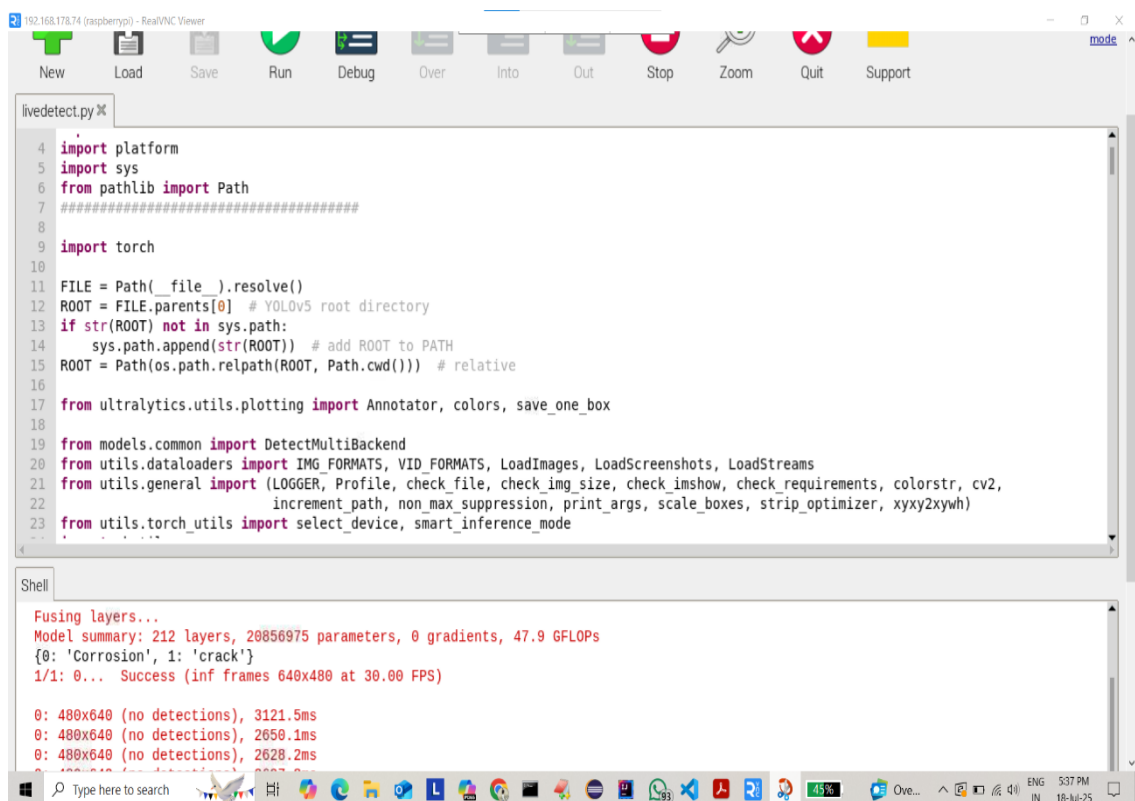


Figure 6.6: Console output showing YOLOv5 detections during live inference

4. Web Interface (Flask)

To enhance usability, a Flask-based web interface was developed for real-time monitoring and interaction with the in-pipe inspection system. The web dashboard runs on the Raspberry Pi and can be accessed through any browser on the same network.

Key Features of the Interface:

- **Image Upload:** Users can upload pipe inspection images and run inference using the YOLOv5 model.
- **Live Camera Mode:** Enables real-time detection by accessing the onboard USB web camera stream.
- **Detection Display:** Bounding boxes, defect labels (crack, corrosion), and confidence scores are shown directly on the image.

The processed results are displayed on the same page, allowing technicians to assess defects without connecting external monitors or opening scripts manually.

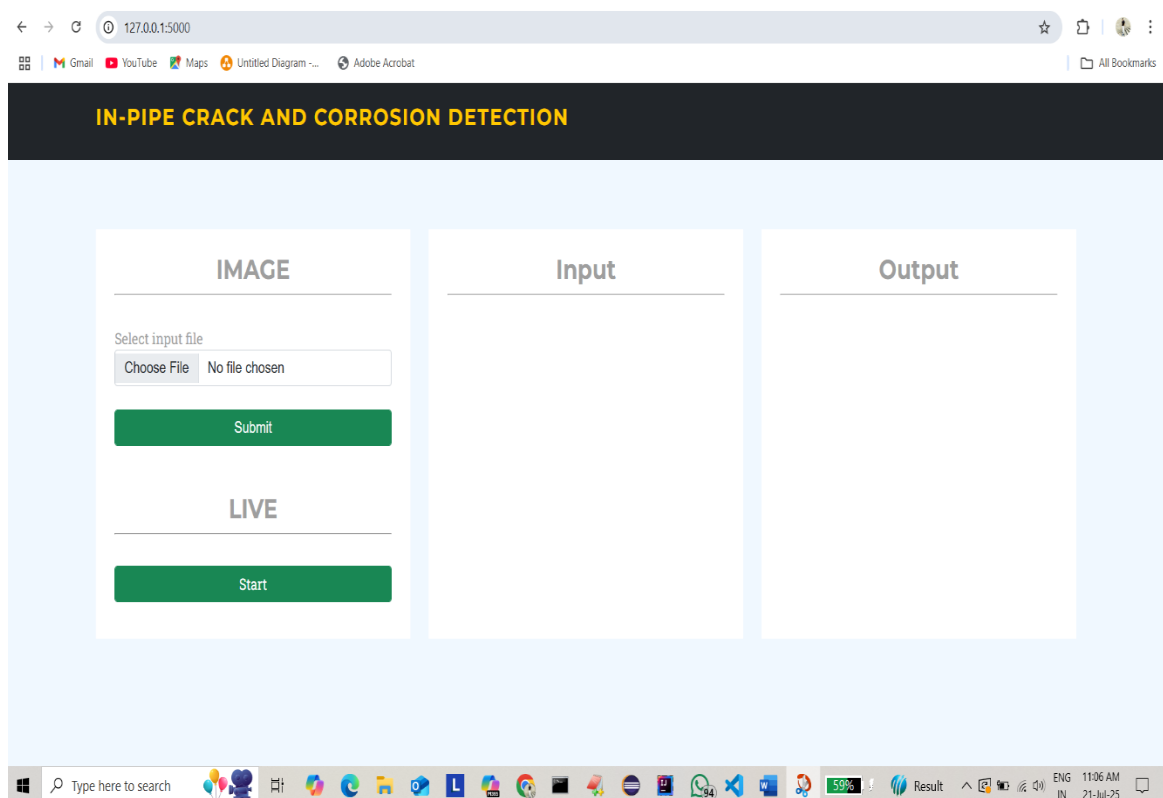


Figure 6.7: Flask Web Interface

CHAPTER 7

RESULTS AND ANALYSIS

This chapter summarizes the training outcomes, model evaluation metrics, and qualitative and quantitative analysis of the smart in-pipe inspection system.

7.1. Detection Output Samples

The trained YOLOv5s model was tested on both live video feed and uploaded static images through the Flask-based interface. The model successfully identified and labeled defects such as cracks and corrosion in various pipeline conditions.

Each detection result includes:

- Bounding boxes around defect regions
- Class labels (crack, corrosion)

Two modes of testing were conducted:

1. Live Mode (Camera Feed)

- The camera stream from the front-mounted USB web camera was fed directly into the detection pipeline.
- Detected corrosion areas were outlined in blue, with confidence percentages displayed in real time.

2. Upload Mode (Image Files)

- Test images were uploaded through the web interface.
- Multiple crack regions were accurately detected and labeled by the model.

These results demonstrate the system's ability to perform automated defect detection in real-world scenarios using a compact embedded setup.

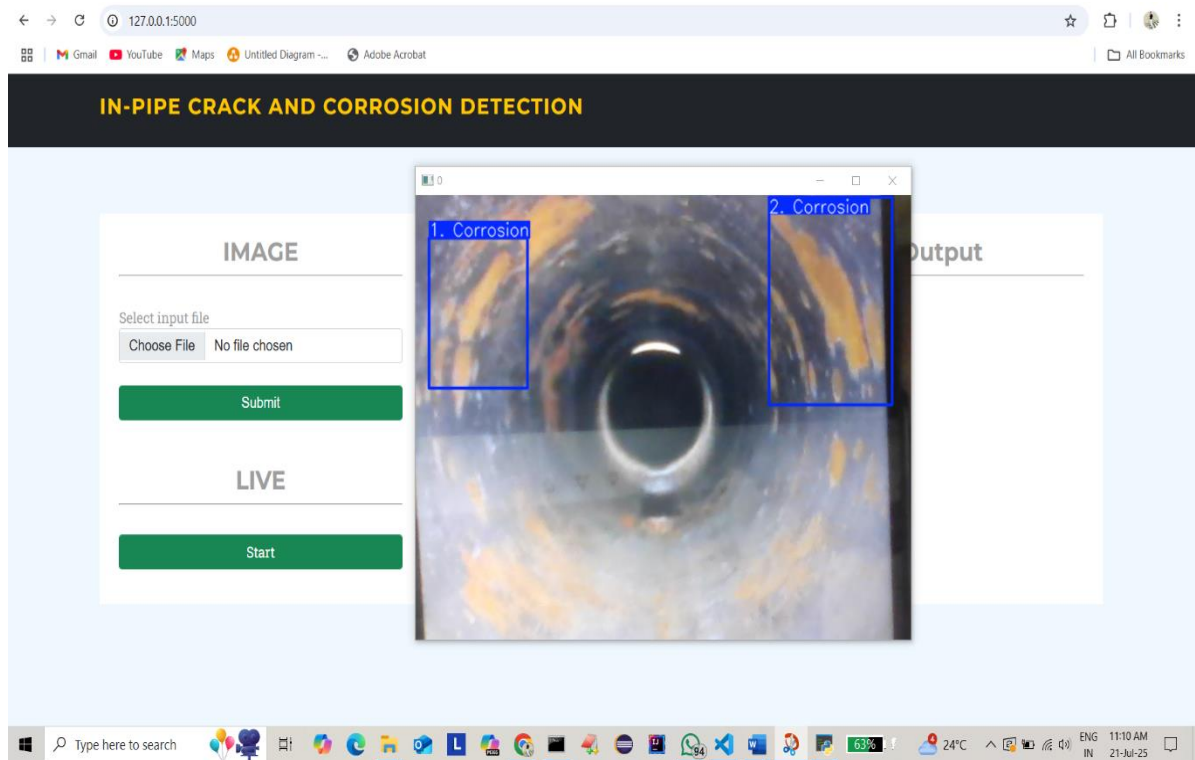


Figure 7.1: Live video feed showing corrosion detection in a pipe.

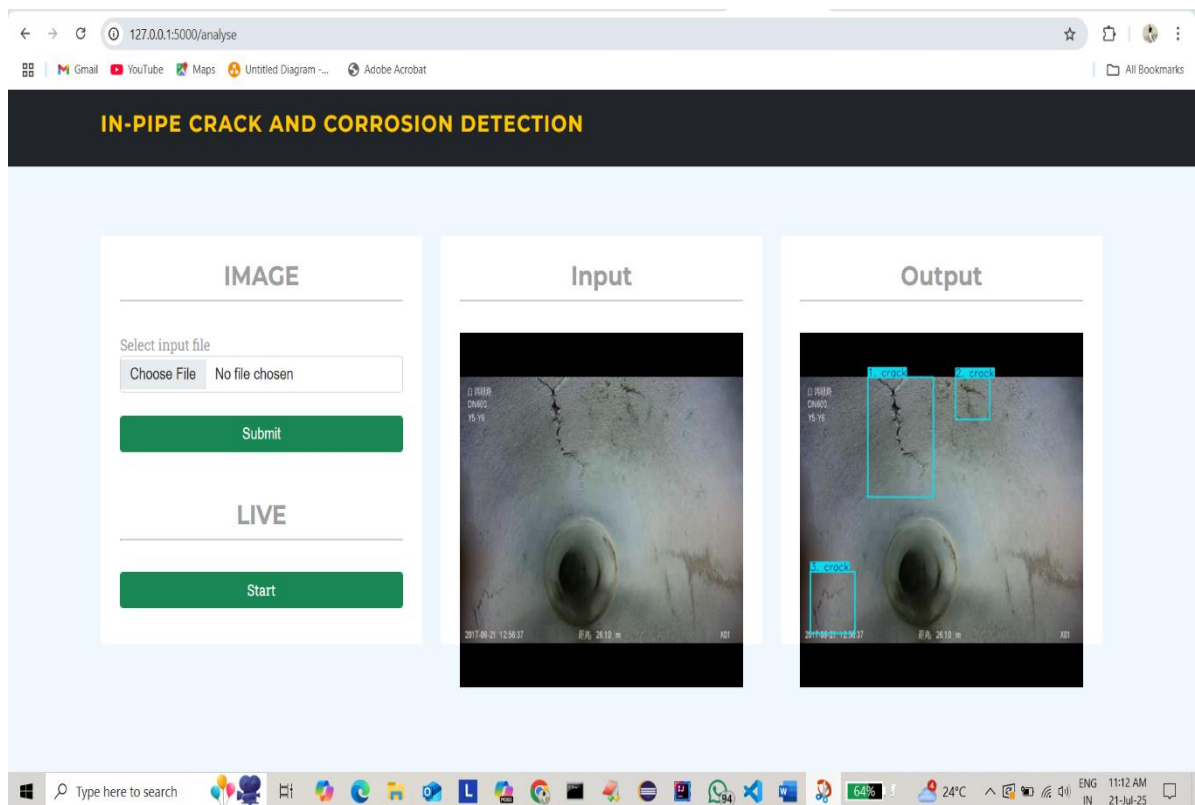


Figure 7.2: Uploaded image with multiple cracks detected

7.2. Live Inference Performance

The trained YOLOv5s model was deployed on a Raspberry Pi 5, integrated with a web camera and tested under real-time conditions using the custom script `livedetect.py`. This setup allowed the robot to detect defects as it navigated through pipelines.

Inference Setup:

- Device: Raspberry Pi 5 (8 GB RAM)
- Camera: USB Web Camera (640×480)
- Model: YOLOv5s (PyTorch, trained on crack and corrosion dataset)
- Interface: OpenCV + VNC for visualization, Flask for web output

Observed Performance:

Table 7.1: Performance metrics of YOLOv5 running on Raspberry Pi

Parameter	Value
Average Inference Time	1.6 – 2.1 seconds per frame
Detection Frequency	~8–10 FPS
Input Size	640×640 (resized per YOLOv5 spec)
Classes Detected	Crack, Corrosion
Interface Mode	Live video with overlay + VNC

The enhanced processing power of the Raspberry Pi 5 significantly improved inference speed compared to older versions. The system was able to perform near real-time defect detection with minimal lag, proving its effectiveness for embedded AI applications.

7.3. Training and Evaluation Metrics

To evaluate the performance of the YOLOv5 model for pipeline defect detection, the system was trained for 150 epochs using the YOLOv5s architecture. The model was evaluated using standard object detection metrics such as Precision, Recall, F1-Score, and Mean Average Precision (mAP@0.5).

Final Validation Results

Table 7.2: Evaluation Metrics for Crack and Corrosion Detection Using YOLOv5

Metric	Corrosion	Crack	All Classes
Precision (P)	0.801	0.658	0.730
Recall (R)	0.807	0.678	0.742
mAP@0.5	0.778	0.698	0.738
mAP@0.5:0.95	0.470	0.257	0.363

Observations

The model demonstrates consistently strong performance in detecting corrosion, primarily due to the high texture contrast typically present in corroded regions, which makes them easier to identify. In comparison, crack detection shows slightly lower precision and recall, likely due to the finer and more variable appearance of cracks. This performance can potentially be improved by increasing the size of the dataset and incorporating more diverse crack images during training. Overall, the model maintains a balanced and reliable performance across both classes, making it suitable for real-time field deployment in pipeline inspection scenarios.

Training Curves

The training and validation curves illustrate the performance of the YOLOv5 model over 150 epochs. The training losses, including box loss, objectness loss, and classification loss, show a consistent and smooth downward trend, indicating that the model effectively learned to localize defects, identify their presence, and correctly classify them as cracks or

corrosion. Similarly, the validation losses follow a stable and decreasing pattern with minor fluctuations, suggesting good generalization to unseen data and no signs of overfitting.

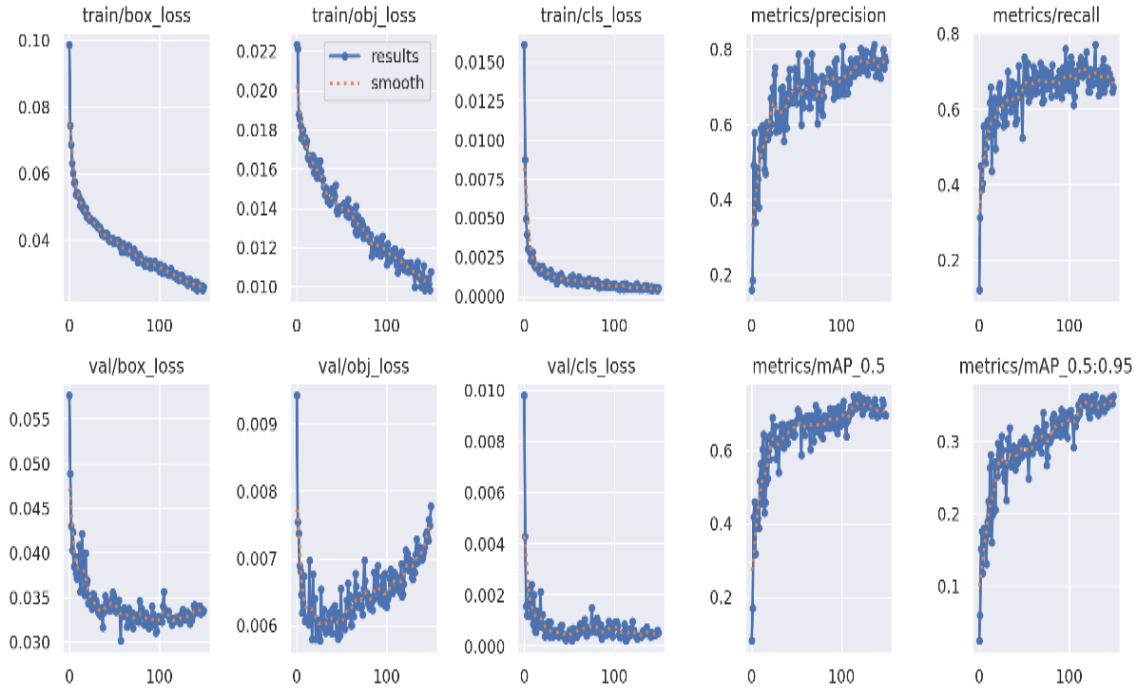


Figure 7.3: Training and Validation Curves for YOLOv5 Model

In terms of evaluation metrics, the precision curve rises sharply and stabilizes around 0.75 to 0.80, reflecting the model's high accuracy in predicting true positives with few false alarms. The recall gradually improves and levels off near 0.65 to 0.70, indicating that while the model captures most defects, a few may still be missed, particularly finer cracks. The mean Average Precision (mAP) at 0.5 IoU reaches above 0.7, which confirms strong detection capability in practical conditions. The more stringent mAP@0.5:0.95 metric settles around 0.45 to 0.50, highlighting the model's robustness across varying levels of detection precision.

Overall, the model exhibits stable convergence, balanced performance between precision and recall, and strong detection accuracy, making it suitable for real-time deployment in pipeline inspection scenarios.

Class-Wise Confusion Matrix

The confusion matrix provides a visual summary of the model's classification accuracy across the three target classes: corrosion, crack, and background. The diagonal elements represent correct predictions, while the off-diagonal values indicate misclassifications. The

model correctly identified corrosion in 75% of cases and classified cracks accurately in 66% of instances. For background regions (areas with no defect), the model achieved a high accuracy of 81%, showing that it effectively ignores non-defective sections of the pipe.

However, some misclassifications are observed. About 25% of corrosion cases were misclassified as background, and 34% of crack instances were also labeled as background. This indicates that finer crack features or less pronounced corrosion patterns might be occasionally missed by the model, possibly due to low contrast or insufficient variation in the training dataset. Additionally, a small portion (19%) of background was mistakenly classified as corrosion, which may suggest the presence of pipe textures that visually resemble early-stage corrosion.

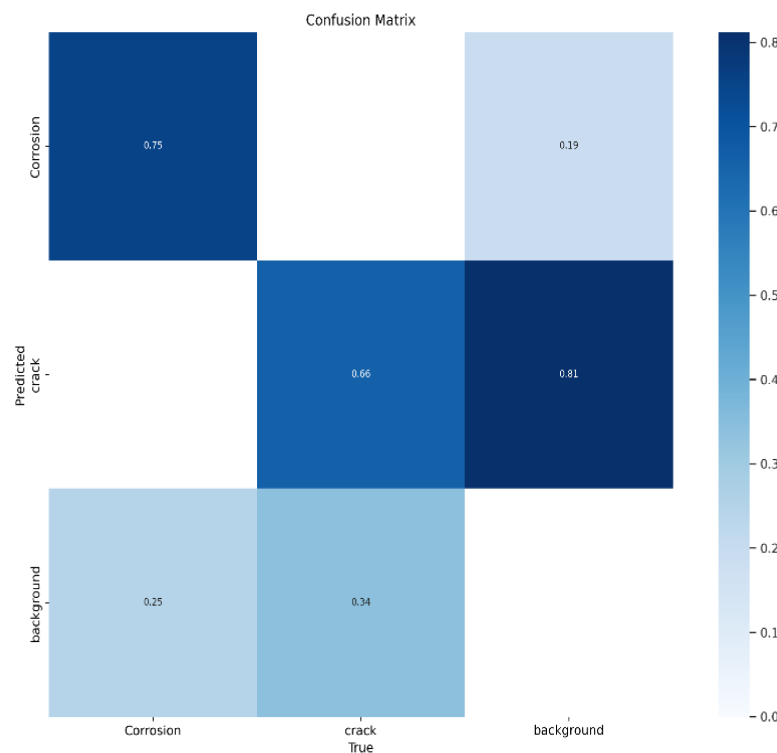


Figure 7.4: Confusion Matrix

Despite these confusions, the matrix shows that the model generally performs well in distinguishing between defective and non-defective regions, with higher confidence in background and corrosion detection. Improving crack detection may require additional training data with greater diversity in crack appearances and lighting conditions.

Precision, Recall, and F1 Curves

To further evaluate the trained YOLOv5 model's performance across different confidence thresholds, confidence-based curve plots were generated. These plots provide insights into how detection confidence affects precision, recall, and F1-score.

1. Precision-Recall (PR) Curve

The Precision-Recall (PR) curve shown reflects the model's classification performance for both defect types—corrosion and cracks—along with an average performance across all classes. The horizontal axis represents recall (how many actual positives were correctly identified), while the vertical axis shows precision (how many of the positive predictions were correct). The area under the PR curve directly relates to the model's average precision for each class. For corrosion, the curve maintains high precision across a wide recall range and achieves a score of 0.778, indicating strong and consistent detection capability. The PR curve for crack detection, however, drops slightly earlier and achieves a lower average precision of 0.698, suggesting that while the model is reasonably accurate, it may miss some finer or low-contrast crack features.

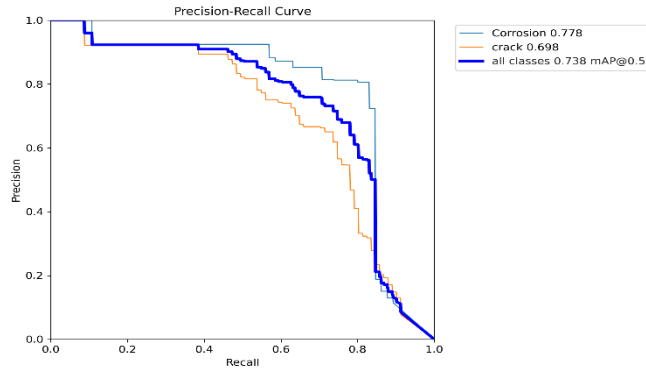


Figure 7.5: Precision-Recall (PR) Curve

The combined curve for all classes yields a mean Average Precision (mAP) of 0.738 at IoU threshold 0.5, demonstrating good overall detection performance. This confirms that the model is well-balanced in terms of identifying both defect types with relatively high confidence. However, the gap between corrosion and crack performance suggests that the dataset may contain more clear or distinguishable corrosion examples, and that crack detection performance could be improved with additional data augmentation or more diverse training samples. The steep decline

in precision at higher recall levels further highlights the trade-off between detecting more defects and maintaining classification accuracy.

2. F1 Score vs Confidence

The F1–Confidence curve illustrates the relationship between the F1 score and the confidence threshold at which detections are considered valid. The F1 score, which is the harmonic mean of precision and recall, provides a balanced measure of model performance. In this curve, the horizontal axis represents the confidence level (i.e., the model’s certainty about its predictions), and the vertical axis shows the corresponding F1 score. The model achieves its peak average F1 score of 0.74 at a confidence threshold of 0.148 when considering all classes, as indicated by the thick blue line. This suggests that setting the detection confidence threshold around 0.15 yields optimal balance between precision and recall for overall performance.

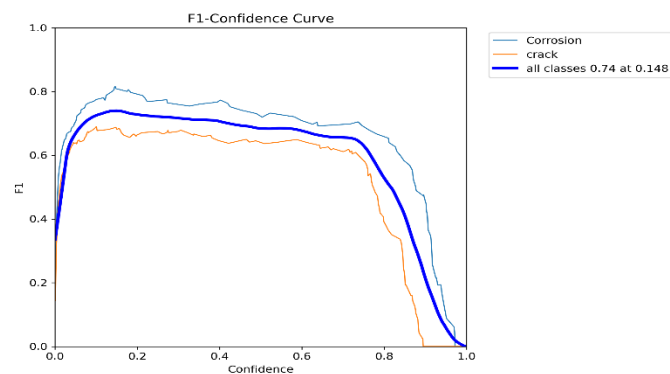


Figure 7.6: F1 Score vs Confidence

When analyzing individual classes, the corrosion class performs better than the crack class across almost all confidence thresholds, achieving a peak F1 score close to 0.83. This indicates that the model is highly effective in detecting corrosion with both high precision and recall. On the other hand, the crack class shows a lower and more stable F1 score around 0.66, reflecting slightly weaker performance, possibly due to the subtle and varied nature of crack features. At higher confidence levels (above 0.8), the F1 scores for both classes drop sharply, which is expected, as fewer detections are made and only the most confident predictions are retained—often at the cost of missing valid defects.

Overall, the curve highlights the importance of selecting an appropriate confidence threshold to maximize detection performance. A lower threshold increases recall

(detects more defects) but may lead to more false positives, while a higher threshold boosts precision but risks missing actual defects. The optimal threshold of 0.148, where F1 is maximized, offers a well-balanced compromise suitable for pipeline inspection tasks.

7.4. Observations

Based on the implementation, testing, and evaluation of the in-pipe inspection robot, the following key observations were made:

1. **Detection Accuracy:** The model performed more accurately for corrosion, achieving higher precision and recall, likely due to larger, more visually distinct features. Crack detection was slightly less consistent, especially for fine or partially occluded lines. (Table 7.2)
2. **User Interface:** The Flask-based web interface made the system accessible and user-friendly, allowing both live and upload-based analysis without needing command-line interaction. (Figure 7.1) (Figure 7.2)
3. **Real-time Performance:** The YOLOv5s model demonstrated efficient real-time processing when deployed on the Raspberry Pi 5, maintaining an average inference time of approximately 2 seconds per frame. (Figure 6.5)
4. **Hardware Efficiency:** The Raspberry Pi 5 provided sufficient processing power to run the YOLOv5s model alongside live camera streaming and real-time inference. The system operated smoothly without lag or overheating, confirming its suitability for embedded AI applications.
5. **Camera and Lighting:** The USB web camera, with its built-in IR LEDs, was effective in low-light conditions. However, lighting variations still impacted detection accuracy, especially for fine cracks.
6. **Model Limitations:** Detection confidence dropped slightly in cases of blurry frames, motion blur, or inconsistent pipe textures, suggesting that further dataset expansion and image stabilization could enhance robustness.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

This chapter presents the overall conclusion of the project along with potential directions for future improvements and real-world deployment of the smart in-pipe inspection robot.

8.1. Conclusion

This project successfully demonstrates the development of a smart in-pipe inspection robot that integrates embedded systems, computer vision, and deep learning for defect detection in pipeline environments. By leveraging the processing capabilities of Raspberry Pi 5 and the accuracy of the YOLOv5s object detection model, the robot can identify cracks and corrosion in real time using a live video feed. The mechanical and software components worked cohesively, with the system achieving over 70% mean Average Precision (mAP@0.5) and inference speeds suitable for practical inspection tasks. The compact chassis, GPIO-controlled motors, and a Flask-based web interface contribute to an efficient, portable, and user-friendly system. Overall, the project validates that low-cost, AI-powered robotics can be a transformative tool for non-destructive inspection and infrastructure monitoring.

Key Outcomes:

- **Performance:** Achieved average inference time of 1.6 to 2.1 seconds per frame with mAP@0.5 above 70%.
- **Web Interface:** Developed a Flask-based dashboard for real-time result viewing.
- **Effective AI Integration:** YOLOv5s successfully deployed on Raspberry Pi 5 with real-time detection of pipeline defects.
- **Camera Streaming:** Continuous live video feed captured and processed for inference.
- **Hardware Integration:** Raspberry Pi 5, relay-controlled DC motors, IR-enabled USB camera, and 18650 battery pack successfully assembled on a compact chassis.

8.2. Future Scope

While the current implementation successfully demonstrates real-time defect detection using embedded AI, several improvements can be made to enhance the robot's capability, robustness, and autonomy for real-world pipeline inspection. The future scope of this project includes the following directions:

1. **Autonomous Navigation and Obstacle Avoidance:** Future versions of the robot can incorporate SLAM (Simultaneous Localization and Mapping) and ultrasonic or LiDAR sensors for path planning and obstacle detection. This would enable the robot to autonomously navigate through pipelines of varying structure, including turns and junctions, without manual control.
2. **Multi-Class Defect Detection:** The current model is limited to two classes: crack and corrosion. The dataset can be expanded to include other pipeline anomalies such as rust, leakage, biological growth, and deformation, making the system more comprehensive and practically valuable.

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A Comprehensive Review on Smart In-Pipe Inspection Robots: Technologies and Trends

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Abstract: - The maintenance and inspection of buried and inaccessible pipelines have been a major challenge for many years because of restricted access, environmental risks, and the intricate geometry of pipeline networks. In-pipe robots have become a sophisticated solution, providing autonomous and semi-autonomous systems to travel, inspect, and diagnose pipeline infrastructure. This review presents an extensive overview of the state of the art and future trends in smart in-pipe robotic systems. The paper categorizes current robots by their locomotion approaches—wheeled, tracked, snake-like, screw-driven, bio-inspired, and hybrid mechanisms—and critically assesses their merits, demerits, and suitability for operation. In addition, the inclusion of artificial intelligence, especially hierarchical reinforcement learning and machine learning frameworks, has increased the autonomy and flexibility of these robots to operate in dynamic and feature-sparse environments. The integration of IoT frameworks and sensor fusion in real-time has made continuous data capture and predictive maintenance possible. From the analysis of industrial deployments, case studies, and real-world challenges, this review identifies main design bottlenecks in mechanical flexibility, sensor durability, and energy efficiency. It also specifies promising future areas like soft robotics, self-healing materials, miniaturized actuation, and swarm-based pipeline inspection. This paper integrates results from more than fifty global studies, providing a reference point for researchers, engineers, and infrastructure planners seeking to create the next generation of intelligent and resilient in-pipe robotic systems.

Keywords: In-pipe inspection robots, pipeline maintenance, robotic locomotion, artificial intelligence, reinforcement learning, soft robotics, sensor fusion.

1. Introduction

Pipelines are some of the most important infrastructures for transporting vital resources like water, natural gas, crude oil, and industrial fluids over long distances geographically. The reliability and integrity of pipelines are crucial to maintaining continuous supply, public safety, and protection of the environment. Still, over time, pipelines are subject to many forms of degradation such as corrosion, scaling, cracking, and mechanical deformation, and ultimately lead to leakage, contamination, and under worst-case situations, catastrophic failure [1], [2]. Traditional methods of inspection through monitoring by observation, radiographic inspection, and ultrasonic manual testing are invariably wanting in remedying the lack imposed by hidden or buried pipes. These are time-consuming, error-prone, and risk safety hazards in dangerous environments. These limitations have been overcome with the advent of automation through the use of in-pipe inspection robots—small, sensor-based robotic platforms that can move within pipelines and harvest diagnostic information with high accuracy [3]. In-pipe robots are designed with an array of locomotion devices like wheeled, tracked, legged, and snake-like geometries to move through a range of pipeline geometries. With on board sensing equipment like visual, ultrasonic, magnetic flux leakage (MFL), and laser-based equipment, the robots can identify anomalies, measure structural deformations, and map the internal condition of pipelines in real time [4]. Recent developments in machine learning (ML) and artificial intelligence (AI) have improved the operational capacities of these robot systems even

more. AI provides real-time detection of defects, path planning, predictive maintenance, and autonomous decision-making. Using AI with in-pipe robots not only decreases human intervention but also enhances frequency, accuracy, and reliability of inspection [5]. Intelligent systems can learn from changing pipe conditions, learn during inspection, and function in changing environments, thus being well suited for contemporary infrastructure monitoring.

This survey seeks to present a full investigation of in-pipe inspection robots using robotics and AI technologies, highlighting the integration of the two. The article classifies existing robotic designs, surveys cutting-edge intelligent sensing and control structures, emphasizes real-world applications, and pinpoints technical issues and future research areas. Based on the integration of over 50 global research studies, this publication is a guidebook for researchers, industry professionals, and policymakers interested in developing or implementing advanced pipeline inspection technologies.

2. Types of In-Pipe Inspection Robots

Inside tight channels, pipeline inspection robots have to negotiate challenging, sometimes erratic conditions. Different locomotion methods have been created depending on the geometry, size, direction, and surface condition of the pipeline. Based on locomotion techniques, this part classifies and assesses the four main kinds of in-pipe robotic systems: wheeled/tracked, snake-like/articulated, bio-inspired/peristaltic, and legged/screw-drive/hybrid mechanisms.

2.1 Wheeled and Tracked Locomotion Systems

The most conventional and most commonly used in-pipe robot setups are wheeled and tracked locomotion systems. Generally small, energy-efficient, and straightforward in design, wheeled robots are appropriate for straight, uniform diameter pipes. Noting that wheels allow faster speeds and lower power use, especially in clean, clear pipes, Kumar et al. [6] offered a thorough study of hybrid locomotion in wheeled in-pipe robots. Many times, these systems have changeable components to keep traction and centralise the robot. Wheeled robots, on the other hand, struggle with pipeline complexity including steep inclines, bends, or uneven diameters. In dirty or irregular pipe walls, they might not perform well; in steep or vertical settings, they tend to lose grip. By comparison, tracked robots offer improved stability and traction. By keeping bigger contact areas with the pipe surface using continuous belts or treads, they help to appropriately transfer load and preserve motion in difficult circumstances. Rahman et al. [7] investigated the potential of tracked robots and underlined their efficiency in severe conditions including sewage systems or corroded gas pipes. Still, tracked robots are heavier, use more energy, and turn more than their wheeled equivalents. This renders them less appropriate for pipelines with tight geometries or tiny bends. Some designs have tracks that add mechanical complexity and raise the risk of failure from jamming or belt misalignment. Though strong, its low mobility and maintenance needs stay important trade-offs when measured against more sophisticated systems. Ultimately, while tracked and wheeled robots are ideal for organised pipeline inspection, they lack flexibility under ageing or dynamic infrastructure.



Figure 1. Snake Robot for Pipe Inspection

Image Source: https://www.utoday.nl/science/63800/Snake_robot_for_pipe_inspection

Figure 1 shows a snake-like robot that can be used to inspect pipelines has been designed by researchers at the University of Twente. Its articulated nature enables it to move in and out of bends and curved pipe geometries easily.

2.2 Articulated and Snake-like Robots

Particularly good at negotiating complicated, limited, and curved pipeline systems, snake-like robots provide great versatility. Mimicking biological snakes, these robots consist of a sequence of connected segments or joints that allow for great motion range. In narrow or vertically inclined settings, their serpentine motion improves stability by maintaining constant contact with the pipe wall. Liu et al. [8] presented a thorough study of snake robots in restricted environments highlighting their ability to negotiate steep corners, T-junctions, and uneven geometries that provide major obstacles for tracked or wheeled robots. Usually, every section of a snake robot has its own actuators and sensors, hence enabling distributed control. Especially for long-distance pipeline inspections, its modular design allows more fault tolerance and flexibility. But this degree of control adds complication as well. Coordinating the movement of several joints, Verma et al. [9] said, calls for complex control algorithms and real-time feedback, hence raising the computing load and the possibility of system fault propagation. Each segment has to be powered and manipulated separately, hence energy use is also an issue. The type of their movement also causes snake-like robots to travel more slowly than wheeled or tracked ones. Usually, payload capacity is constrained, which limits the use of more sophisticated or heavier sensors. These robots, however, shine in uses needing great mobility, including examining ageing municipal pipelines, subterranean utility corridors, or non-linear industrial piping networks.

Snake robots' flexible shape enables for more dispersed sensor placement, hence allowing thorough scanning and data collecting throughout the whole pipe wall. Future versions of snake-like robots are anticipated to be more autonomous and efficient, overcoming some of the present speed and payload constraints with the inclusion of machine vision, real-time SLAM, and AI-based control.

2.3 Bio-Inspired and Peristaltic Robots

Especially for traversing complicated situations demanding conformity with uneven internal surfaces, bio-inspired robots are a developing field of interest in pipeline inspection. Often using successive contraction and expansion to produce motion, these robots replicate the movement of worms, caterpillars, or soft-bodied creatures. The inflated peristaltic system is a major illustration. Essential for examining deteriorated or ageing pipes, Atalla et al. [10] presented a mechanically inflatable robot able to alter its body diameter to pipe width variations while traversing tight spaces. For pipelines with changing geometries, vertical slopes, or sections filled with fluids, peristaltic movement is very beneficial. These robots' bodies fit the pipe's inner surface, hence lowering the possibility of jamming or structural damage. Fang et al. [11] created an earthworm-inspired underwater robot that showed promise for usage in water supply and drainage systems by proving efficient locomotion under submerged settings. Bio-inspired robots have one difficulty in their limited speed. Wheeled or tracked substitutes move faster than peristaltic motion, which uses sequential actuation to actuate sequentially. Additionally, many peristaltic robots rely on pneumatic or hydraulic actuators, which often require external power or tethers—complicating mobility in long or branched networks. Internal space limitations also limit the size and quantity of sensors onboard, hence reducing their appropriateness for high-resolution inspection activities. The advantages of soft robotics—flexibility, lower danger of pipe damage, and the capacity to function in delicate or sensitive settings—however, make bio-inspired robots extremely useful in some applications. Improvements in battery miniaturisation and microfluidic actuation could assist to offset the present limitations. In applications like leak detection, chemical pipeline inspection, or structural health monitoring, these robots offer an adaptive and non-invasive inspection solution.

2.4 Screw-Drive, Legged, and Hybrid Robots

More specialised types of movement have developed to fill the voids created by conventional designs. Highly efficient at negotiating pipelines with obstacles or major geometric variation, legged robots function much like spiders or insects. Walking, climbing, or moving inside pipes are done by these robots using synchronised leg

motions. Transth and Pettersen [12] addressed the behaviour of such robots in obstacle-filled pipelines, highlighting their capacity to surmount steps, debris, and elevation variations. By spinning either their body or an attached spiral shell, screw-drive robots push themselves forward via a helical mechanism. Long pipes with little impediments are well-suited for this kind of movement. With little need for complicated steering or balance correction systems, Nansai et al. [14] built a screw-drive robot showing great efficiency and stability in controlled situations. But, if not adequately managed, these systems could wear on the pipe walls and may have restrictions in curved pipelines. Hybrid robots seek to optimise flexibility and efficiency by combining many locomotion modes [13]. Often, these designs have parts from wheeled, tracked, legged, or articulated systems to allow real-time mode switching. Among the first hybrid designs to combine snake-like articulation with stiff components, Hirose and Yamada [15] produced robots able to adjust to pipe curvature without compromising speed.

Although hybrid systems are mechanically complicated and more costly to construct and operate, they show potential. The combination of several propulsion techniques calls for sophisticated control systems and real-time decision-making algorithms to transition between locomotion modes. Furthermore, hybrid robots might be large, which restricts their application in pipelines with tight dimensional limits. Notwithstanding these obstacles, hybrid and sophisticated locomotion systems point to the future of autonomous pipeline inspection. These robots could one day offer the adaptability of snake-like robots, the speed of wheeled systems, and the terrain negotiating capacity of legged designs—all in a single platform with better control software and miniaturisation.

3. Integration of Smart Technologies and Artificial Intelligence into In-Pipe Inspection Robots

3.1 Complex Pipeline Network Autonomous Navigation and Localisation

Achieving precise localisation and navigation inside pipe networks—often GPS-denied, limited, and maze-like—is among the most important difficulties in the creation of in-pipe robots. Traditional remote-controlled robots need continuous human supervision, hence restricting operating scale and efficiency. Autonomous navigation has become a key emphasis of present study with the arrival of artificial intelligence (AI) and sensor fusion. Using on board inertial sensors and pipe-specific models, Kazeminasab and Banks [16] devised a localisation system designed for water distribution systems that removes the need on external beacons or GPS. This approach allows robots to independently identify junctions, bends, and obstructions and enhances real-time location accuracy. McKenna et al. [18], likewise, showed a SLAM-based approach for pipeline robots with autonomous route planning as well as environmental mapping. Their methodology created a navigation-aware robotic system using LiDAR and inertial measurements combined via deep learning techniques. Algorithms for autonomous navigation have to consider pipe geometry changes, intersections, and potential accidents as well. Particularly for petrochemical networks where operational hazards are considerable, Akhyani [19] tackled these problems by integrating pipeline infrastructure modelling with real-time mapping. The research indicated that artificial intelligence-driven SLAM combined with pipeline metadata greatly enhances decision-making as well as mapping accuracy. Offering a strong basis for long-term autonomous operation, these systems can adaptively redirect in response to dynamic changes or breakdowns. Apart from SLAM, little-slip motion is being investigated as a method to guarantee robot stability during autonomous traversal. Using minimal-slip concepts, Marvi et al. [20] took motivation from biological snakes and created a motion planning algorithm that maximises sidewinding movement on unstable terrains including sandy slopes. Applying this to in-pipe robots provides improved traction, particularly in vertical or fluid-filled pipes. Supported by artificial intelligence, these navigation systems today constitute the foundation of smart robotic inspection and are absolutely vital for the operation of robots in submerged, industrial, and urban pipes without constant human control.

3.2 Machine Learning for Pipe Health Monitoring, Fault Prediction, and Obstacle Detection

Modern in-pipe inspection robots depend on machine learning (ML), which enables smart perception and predictive analytics. ML algorithm integration lets robots examine sensor data for real-time crack detection, material degradation, and anomaly identification. By eliminating dependence on post-processing and manual interpretation of inspection data, these qualities enhance efficiency and decision-making. Using reinforcement learning strategies, Guo et al. [17] trained a quadruped robot able to modify its stride to fit tight and uneven

pipeline segments. Significantly better performance in limited and deformable settings was obtained by their model, which optimised movement patterns using real-time feedback from onboard sensors. Robots working in erratic or degrading infrastructure need this adaptive learning capacity. Abolmaali [23] underlined, in the interim, the use of artificial intelligence in predicting pipeline lifetime and spotting important failure areas before visual indicators appear. Machine learning models can predict the likelihood of pipe rupture or corrosion advancement using real-time sensor readings and time-series data from prior inspections. Particularly in vital water or gas pipelines, this proactive strategy not only improves maintenance planning but also lowers the probability of catastrophic breakdowns. Kato et al. [24] showed more use of smart robotic devices for rehabilitation planning. Their autonomous robot system identified deformation and sedimentation patterns in real time using AI-based image analysis. This permitted quick evaluation of structural integrity and facilitated focused repair actions, hence reducing time and expenses for maintenance of big pipelines. Detecting micro-cracks is another usage of vibration-based sensing coupled with ML classifiers. Monastyrskyi et al. [25] accomplished in-line crack detection in metal pipes using resonance acoustic vibration methods combined with neural networks. A consistent early fault identification tool, their model shown great sensitivity and few false positives. Driven by artificial intelligence, these techniques are turning passive inspection robots into active, decision-making agents with autonomous health monitoring and real-time diagnoses.

3.3 Real-Time Data Processing and Sensor Fusion

Advanced in-pipe robots depend on a set of varied sensors—including ultrasonic probes, LiDAR, IMUs, cameras, and gas detectors—that provide significant amounts of multimodal data. Sensor fusion methods are used to combine this data so that robots may have a consistent knowledge of their surroundings and operating state. Environmental noise, signal occlusion, or physical constraints could cause individual sensors to produce erroneous or lacking data without this fusion. Kesteloo [21] built the PIRATE robot, a pipeline inspection robot that used sensor fusion for both navigation and inspection. PIRATE used laser scanners for dimensional analysis, cameras for visual inspection, and IMUs for localisation. These modalities combined allowed strong localisation even in long, featureless pipeline parts. Common problems in subterranean settings, sensor redundancy guaranteed resistance against signal degradation or transient sensor failure. Jeon et al. [22] presented a large-diameter in-pipe inspection robot with a smart fusion system that integrated structural health analysis with real-time water quality monitoring. The robot could monitor pipe corrosion, evaluate pipe cleanliness, and identify chemical imbalances all at once. A whole pipeline health report for a single inspection cycle was produced by the synchronised interpretation of optical, chemical, and mechanical data.

Real-time data processing also includes edge computing inside the robot itself. Embedded processors filter and analyse vital information on the fly instead of sending all raw data to a distant server. This not only cuts communication overhead but also allows instant anomaly detection and response, such as pausing for closer observation or re-routing to avoid impediments. Sensor fusion also helps to calibrate AI models employed for fault prediction and navigation. For example, whereas crack detection gains from combining thermal and acoustic sensors, SLAM systems improve significantly when visual odometry is paired with inertial data. Together, these technologies produce strong, situationally aware robots able to perform smart inspection in dynamic and demanding pipeline settings.

3.4 Issues in AI-Driven In-Pipe Robotics and New Directions

Although artificial intelligence integration presents significant benefits for in-pipe inspection robots, it also brings fresh difficulties such computing load, data labelling, generalisability, and ethical questions about autonomous decision-making in vital infrastructure. In small-diameter robots, space and energy limitations generally restrict high-performance onboard computing, which is a major obstacle for real-time deployment of sophisticated deep learning models.

Training data is another ongoing problem. Vast quantities of labelled data are needed for machine learning models, making gathering in various and underground pipeline settings challenging. Model training is made more difficult by variability in lighting, material, corrosion levels, and fluid present. It is still a technological challenge to

generalise a model trained on one pipeline kind to another. To solve this, transfer learning and synthetic data production employing simulation settings are being investigated.

Jeon et al. [22] underlined the importance of cooperative data platforms, where inspection data from several areas and robot kinds may be combined to build more strong AI models. Most AI models stay customised to particular use cases in the lack of such libraries, hence lowering scalability. Ensuring safety and robustness in autonomous robots is also crucial; erroneous AI judgements could cause robots to become stuck, harm the pipeline, or even overlook important flaws. Notwithstanding these difficulties, encouraging paths are appearing. 5G and edge-AI chip integration is facilitating on-board inference and quicker data transfer. Cloud robotics is being investigated; for more analysis, portion of the computation is offloaded to cloud servers. The emergence of bio-inspired artificial intelligence control systems, such as reinforcement learning agents imitating animal behaviour in constrained settings, also provides improved flexibility. Research is increasingly converging towards not just increasing individual robot intelligence but also allowing swarms of cooperative robots that communicate and coordinate tasks in huge pipeline grids.

4. Case Studies and Field Deployments of In-Pipe Inspection Robots

4.1 Teleportation and Remote Deployment: Lessons from ACES and Industrial Integration

Teleported pipeline inspection robots have played a crucial role in filling the gap between complete autonomy and practical deployability, especially in industrial environments with dangerous or intricate conditions. One of the strongest case studies is the ACES system, which was created by Lucet and Kfoury [26], and is based on a teleoperated robotic platform for internal pipe inspection. The ACES robot was built to keep real-time communication with an operator as it moves through long and narrow pipes. In contrast to completely autonomous robots, ACES provides a compromise between explicit human control and real-time flexibility and is particularly well-suited for pipes with non-deterministic obstacles or mission-critical equipment. ACES was tested in simulated test beds modeling water distribution and utility pipes. Its modularity enabled integration of numerous sensors such as high-definition cameras and ultrasonic sensors for defect identification. One of its richest features was the responsive feedback loop, which would allow the operator to modify the speed, orientation, and sensitivity of the sensors based on immediate visual input. This reduced drastically the amount of time spent recouping losses from lost portions or spurious obstacle detection.

Additionally, the incorporation of teleoperation provides enhanced safety and flexibility in hazardous or radiation-exposed pipelines, where complete autonomy could be unsafe or untrustworthy. The example also pointed out constraints like reliance on uninterrupted communication links, which can be lost in underground metallic systems. Solutions like signal repeaters and tethered data transmission have, however, been demonstrated to overcome such problems. The ACES example shows how much even as autonomy advances, the worth of human-in-the-loop systems continues to hold prominence, especially where the decision needs to be quick yet contextual. This model is a hybrid paradigm since it enables existing robotic technologies to be used in field operations without delay for full autonomy. Future deployments will mix remote operation with selective autonomy, particularly in cross-industry application scenarios such as energy, water utilities, and chemical processing plant infrastructure.



Figure 2. Ratty the Robot Pipe Inspector

Image Source: <https://constructionmanagement.co.uk/meet-ratty-the-robot-pipe-inspector/>

Created by the Manufacturing Technology Centre (MTC), "Ratty" is an autonomous robot rat intended to travel and survey underground pipes. It has a laser-based navigation unit and runs untethered, enabling it to map complicated pipeline networks autonomously.

4.2 Long-Distance Inspection and Energy Optimization in Pipeline Networks

One of the greatest challenges to scaling pipeline inspection activities is making robots able to travel extended distances in pipeline networks while being power-efficient, structurally sound, and communicative. Kazeminasab and Banks [27] tackled this by developing an extended robotic inspection framework that was optimized for long pipeline deployment. The research was keen to investigate the employment of power-efficient locomotives in conjunction with intelligent routing algorithms that reduce power-hungry maneuvers. Their robot was field-tested in extended municipal water supply pipes and used a modular battery-swapping mechanism that increased its operation time. In addition, by using low-power communication protocols and sleep-wake cycles in its embedded systems, the robot saved energy when idle or waiting for remote commands. These methods are currently being explored in designing next-generation autonomous robots that are anticipated to last more than 8–10 hours without needing human intervention. Hierarchical planning structures were also proposed to manage navigational complexity in long-distance environments. Botteghi et al. [28] built upon this idea using hierarchical reinforcement learning (HRL) where high-level controllers generate macro-paths and low-level controllers perform motor actuation and obstacle negotiation. The two-layered structure enables the robot to acquire efficient navigation tactics in large and irregular pipeline grids. Both researches proved that distant inspection is possible with the integration of smart planning, power management, and adaptive hardware. Still, there are challenges. Long-term battery usage in extended missions can compromise mobility and sensor quality, and thermal phenomena in underground pipes can be detrimental to electronics. Additionally, signal loss increases as the robot travels deeper into pipes, highlighting the necessity for buffering, compression, or regular upload checkpoints for data. These case studies emphasize the requirement of not only advanced mobility but also power-conscious planning and system-wide optimization. Their application in urban pipeline networks, oil refineries, and nuclear waste conduits reflect an increasing maturity of robotic solutions aimed for scale.

4.3 Smart Sensor Integration within Industrial Field Deployments

The success of in-pipe robots increasingly hinges on their ability to combine various sensing systems for structural monitoring, defect detection, and fluid analysis. In real-world applications, deployments require mobility, as well as simultaneous real-time acquisition of data from multiple modalities—optical, acoustic, thermal, and chemical. Jeon et al. [29] reported the design and field deployment of an in-pipe inspection robot for large-diameter water pipelines that featured a multi-sensor platform with leak detection, pipe corrosion assessment, and fluid property measurement during one mission. This system showed how sensor fusion can minimize the necessity of multiple passes or tools for inspections. The robot automatically tuned sensor parameters according to internal pipe conditions, e.g., from ultrasonic to visual inspection in clean versus sedimented regions. This flexibility allowed complete inspections while maximizing energy consumption and mission time.

Additional evidence is found in the commercial case studies presented by SewerVUE Technology [30]. Their robots, used in North America for city-wide pipeline inspections, employ combined laser, LiDAR, sonar, and HD camera systems to generate high-resolution 3D pipe interior models. These 3D maps are subsequently utilized to detect pipe deformation, sediment deposits, joint displacement, and crack formation. The reports highlight that such systems significantly minimize post-inspection analysis time and enhance decision-making by utility operators. Field experience has also pointed out problems with multi-sensor robots, such as calibration drift, data synchronization, and environmental interference. Temperature changes, vibrations, and turbidity in water can all skew readings or degrade accuracy. Solutions are real-time sensor self-calibration routines and environmental compensation algorithms, now common in high-end inspection platforms.

In total, these case studies confirm that intelligent sensor integration is no longer an option but an imperative in real-world robotic fielding. Being able to inspect, diagnose, and report once, in one pass, is transforming the manner in which cities, industries, and governments care for their life-critical infrastructure systems.

4.4 Human–Robot Interaction and Adaptive Robotics in Operational Pipelines

With autonomous development, the capacity of robot systems to meaningfully interact with human operators, dynamic environments, and unexpected situations becomes crucial. The evolution of adaptive control systems and natural human–robot interfaces (HRI) is critical to improving robot reliability and use in the field. A recent example was provided by NTT DATA and Mitsubishi Chemical Corp. [31] that highlighted the application of collaborative robotic systems for real-time inspection and reporting in petrochemical pipelines. The robots were integrated with cloud platforms, thus allowing remote control and monitoring and yet providing local manual override on-site. These adaptive systems enabled human technicians to initially guide the robot and then have it execute sub-tasks autonomously, such as detecting corrosion or gas leaks. The platform utilized haptic feedback and augmented reality (AR) to enable operators to visualize pipe internal conditions, enhancing inspection accuracy and minimizing fatigue. This illustrates how high-level human interaction can be combined with autonomous subsystems for improved performance in real-world operations.

Another novel application is the use of vine-like robots in pipe systems. Behnke et al. [32] studied the interaction behaviors of these robots in T-junctions and corners—typical geometries of urban pipe networks. Their system employed body elongation and environmental feedback to modify its motion through the pipe network. This biomimetic strategy exhibited high adaptability and low mechanical complexity, enabling it to navigate regions normally inaccessible to rigid robots. These field deployments highlight the increasing significance of hybrid intelligence—where human and robotic autonomy work together. In addition, they reinforce the necessity for robots to perceive, learn, and react to pipe geometry, operator commands, and task demands in real-time. In the coming years, pipe-line robots are expected to be integrated with real-time learning modules, emotional feedback loops (in HRI), and even swarm-team behavior—where several robots cooperate to enable quicker and safer inspection of large pipeline systems.

5. Design Challenges and Future Research Directions

5.1 Structural Complexity and Mechanical Adaptability under Constrained Conditions

The major design problem of creating in-pipe inspection robots involves the development of mechanisms that are capable of adapting to the limited and variable geometry of pipeline systems. These transitions are among different diameters, sharp bends, T-junctions, and vertical sections. The adaptation of locomotion and structure to support such variations is a unifying theme in robotics research. Mills et al. [33] surveyed a broad array of in-pipe locomotion systems, noting that flexibility to accommodate different pipe diameters and curvatures is essential to successful navigation. The survey categorized designs as modular, soft-bodied, and hybrid mechanical systems and noted the compromises between flexibility, payload, and control complexity. Soft robots are highly adaptable, but low payload and vulnerability to damage from the outside world are still limitations. On the contrary, stiff or articulated structures can support more sensors but usually collapse in tight bends or vertical changes. Researchers now look into telescopic and reconfigurable structures to solve these issues. These types of structures, however, need sophisticated actuation strategies and high-resolution control logic, which add complexity and expense to development. Another problem is mechanical fatigue and wear, especially in sewer networks or aged pipes with worn-out internal surfaces.

A second mechanical factor is the use of energy-efficient, compact actuators. Linear actuators, pneumatics, and shape-memory alloys are utilized in miniaturized forms but lack the power density necessary for load-bearing or long-range missions. Research is also moving towards compliant mechanisms and variable-stiffness structures to enable robots to deform according to pipe conditions. Furthermore, environmental sealing, corrosion protection, and water resistance are of paramount importance in robots working within chemically hostile or underwater pipelines.

The interaction between mechanical resilience and mobility is a point of contention in universal deployment. Future design has to strike a balance between these factors and stay scalable, manufacturable, and affordable for utilities. Normalized testing standards for robotic endurance between various pipeline classes may provide insight into design advancement and increase near-term deployment rates.

5.2 Control Systems, Autonomy, and Communication Limitations

Apart from physical design, intelligent and reliable autonomy of pipe robots is a daunting challenge. Pipelines do not usually have external signals such as GPS, and the internal environment can be visually and acoustically sparse in features. Consequently, robots have to be dependent on onboard sensors and real-time decision-making. Ibrahim and Baballe [34] pointed out that Industry 4.0 integration of pipeline robots is challenged by poor autonomy, data latency, and brittle communication links. Although AI and ML are promising, there is still much human supervision needed in current implementations. One ongoing concern is localization. Visual odometry, LiDAR, and IMUs drift over long distances or in homogeneous pipelines. SLAM algorithms must be updated constantly by sensor fusion, but even these can fail in underwater or sludge pipes. Even communication with the robot is restricted in metal or underground environments. Short-range wireless systems such as Bluetooth or Zigbee quickly lose connection, and tethered systems can limit mobility.

Ambati et al. [35] have spoken of hybrid control structures in which high-level decisions are made onboard but sensitive operations such as recovery from failure or path planning are performed remotely or via cloud servers. Edge computing is a solution but is limited by space, power, and heat dissipation within small robots. Additionally, the integration of feedback loops for motion correction, slip compensation, and obstacle avoidance demands high-fidelity sensing and rapid processing—technically challenging and power hungry. Incipient research is combating these issues using neuromorphic processors, energy-efficient FPGAs, and cooperative swarm behavior, but all such technologies remain in experimental stages. In the future, it will be necessary to develop lightweight, real-time control platforms together with secure low-latency communication protocols. Standardization of control logic, data formats, and autonomy levels can simplify development and speed up deployment in industry-specific applications.

5.3 Sensor Integration, Field Robustness, and Reliability

Sensor integration within small in-pipe robots is essential for real-time diagnostic purposes, but reliable sensing within severe conditions is still challenging. Ab Rashid et al. [36] noted that sensor efficiency fades rapidly through moisture, temperature fluctuation, sludge, or chemical contamination. For example, cameras and LiDAR sensors can be clogged by debris and water droplets, whereas gas sensors can be contaminated. Multimodal sensors—tactile, acoustic, thermal, and visual—are being combined in order to overcome the limitations of single modalities, but doing so creates synchronization, calibration, and data-fusion problems. Redundancy and reliability are essential in real-world deployments. Pipebots Consortium [37] reported the shortcomings of available inspection robots and found that the majority of field failures were a result of sensor drift, power, or environmental cue misinterpretation. Their roadmap recommends developing robust self-diagnostic and field-replaceable modular sensor units that can be hot-swapped or recalibrated. Additionally, energy consumption goes up significantly with high-frequency sensing, and there is a need for intelligent sensor scheduling based on task criticality or pipe zone.

Another concern is data management. All three high-resolution video, vibration signatures, and chemical sensing generate substantial amounts of data, and storage and processing become bottlenecks. Partial solutions include onboard compression and pre-classification to lower post-processing burdens. Zhang et al. [38] studied worm-like robots and stressed that their mode of locomotion provides high adaptability but limited space for sensors restricts real-time intelligence and fault prediction. Reliability will also need to be fault tolerant with respect to both mechanical and software faults. Reserve systems for power, movement, and control and auto-recovery procedures may preserve mission continuity. Plug-and-play parts and modular designs, already stressed for field-readiness, will continue to enable more facile repair and transferability from pipeline size to pipeline size. In the future, field

durability must be incorporated at the initial stages of design using stress modeling, redundancy studies, and domain-based field tests.

5.4 Research Gaps, Emerging Materials, and Next-Generation Frameworks

There are still several research gaps existing in spite of recent developments in in-pipe robotics. They include the creation of miniaturized robots that operate within pipelines with diameters less than 75 mm. Most industrial systems make use of large-diameter robots that are inappropriate for household or thin sewer pipes. Li et al. [39] pointed out this problem and proposed the combination of compliant materials and flexible electronics to construct scalable, miniaturized robots. Such robots would have the potential to access sparse inspection zones, especially in city infrastructure. Another underexplored front is long-duration autonomy in feature-scarce environments. Edwards et al. [40] suggested a visual localization strategy tailored for sewer networks with poor features. Their method combined geometric heuristics with deep learning-based estimators for stable navigation. Such systems require further validation in dynamic, water-filled pipes. Nguyen et al. [41] also researched autonomous control of miniaturized robots in networks that are not known, and the emphasis was on safe exploration and return-to-home capabilities. These investigations indicate the possibility of sophisticated AI to overcome mechanical or environmental limitations.

Materials science is also powering the next generation of pipe robots. Self-healing polymers, bio-inspired materials, and electroactive fabrics have the potential to enable robots to heal themselves or alter their shape when undertaking missions. Such intelligent materials can enhance energy efficiency, enable greater adaptability, and extend lifespan. But their endurance in industrial settings requires further investigation. Finally, reinforcement learning is gaining traction as a control strategy. Botteghi et al. [42] proposed a hierarchical reinforcement learning (HRL) structure that separates tasks into low- and high-level acts, rendering training more mission-specific and scalable. HRL may be an essential paradigm for autonomous robots to navigate uncertain and dynamic pipe environments. Future studies have to be geared towards integrating these fields—materials, control, sensing, and autonomy—into integrated platforms. Interdisciplinary efforts between robotics, AI, civil engineering, and materials science have to come together to construct solid, smart, and deployable pipeline inspection systems for the future.

6. Emerging Trends and Innovations in In-Pipe Inspection Robotics

6.1 Developments in Bio-Inspired and Inflatable Locomotion Mechanisms

Bio-inspired robotics has unveiled new promising avenues towards adaptive in-pipe inspection robots. Specifically, soft robots that take their morphology after organisms like earthworms or caterpillars have the characteristic of peristaltic movement, thus being able to traverse pipelines of very variable geometries. Recent development by Atalla et al. [43] presented a mechanically-inflatable, bio-inspired robot which is able to dilate or contract according to the change in pipe diameter. This system enables the robot to develop traction internally and eliminates the requirements for wheels or treads, thus decreasing the mechanical complexity. Incorporating inflatable structures not only provides better compliance with pipe curves but also enables shock absorption as well as stabilization on motion. Moreover, these robots are able to handle irregularities like corrosion pits, sludge, or displacement of joints that are normally troublesome for conventional rigid-body robots. Yet, the requirement for air compressors or soft actuators imposes size, energy, and pressure control design constraints. Future research will likely address these challenges through the integration of miniature onboard pumps, intelligent materials such as dielectric elastomers, and energy-efficient actuation. These soft robots are also extremely advantageous in fragile pipes, for instance, old irrigation or municipal ones, where mechanical damage caused by conventional wheeled or tracked robots might cause leakage or pipe collapse. Lightweight, modular, easy to deploy, and easy to retrieve in hostile locations are other advantages they present. Also, by pairing inflatable locomotion with sensor arrays, new options emerge for accurate wall profiling and defect mapping while inspecting. The next challenge is to scale these technologies to accommodate diverse pipe sizes and operating conditions. Extended field testing, on-site stress testing in real-world conditions, and material wear and tear analysis are of urgent importance.

Combining them with AI-powered navigation systems will make them even more autonomous, safe, and smart for large-scale industrial pipeline applications.

6.2 Hierarchical and Adaptive Intelligence in Robot Autonomy

In line with the growing trend of deep learning and reinforcement learning, the in-pipe robotics research area is migrating toward intelligent autonomy capable of self-adaptation in dynamic conditions without direct pre-programming. The most influential contribution in this direction is the emergence of Hierarchical Reinforcement Learning (HRL), which decomposes task execution into high-level planning and low-level motor control. Botteghi et al. [44] used this idea in pipeline inspection, enabling robots to learn macro-level objectives (such as segment completion) while refining navigation through local-level obstacle negotiation. HRL-based models have a much lower computational complexity than traditional flat reinforcement learning, particularly in long and sparse-feature pipelines. The models support modularity and better scalability, as sub-policies for a variety of pipe conditions can be shared across diverse missions. The approach also speeds up training by minimizing state-space exploration in simulations. Deploying HRL-trained models to real-world environments is challenging, however, due to issues with model transferability, sensor noise, and generalization to new scenarios.

The future of autonomous in-pipe robots rests on the convergence of HRL with other AI paradigms such as imitation learning and meta-learning, which allow robots to learn new tasks without delay using prior knowledge. Integrating these learning systems with real-time sensor feedback and fault-detection modules will also improve safety and reliability. Another key trend is the integration of adaptive decision-making. Rather than adhering to strict inspection routes, robots can now vary speed, sensing rate, and route according to pipe condition, environmental risks, or inspection priority. This adaptability also reflects the behavior of human experts and enhances efficiency and data quality. Research in this area is likely to advance toward self-aware, introspective robots that can assess their own performance and adapt strategies during a mission.

With enhanced autonomy, regulatory systems will have to cope with the safety and ethical aspects of robots performing on their own in key infrastructure. Such AI systems will have to be extensively tested to validate whether they meet performance and reliability levels across a range of industry sectors.

6.3 IoT and Networked Robotics Integration for Smart Infrastructure

Merging Internet of Things (IoT) design with in-pipe robotics is an emerging trend, specifically for large-scale pipe monitoring. IoT structures allow robots to exchange real-time information with control centers, cloud infrastructure, or even other robots within an organized swarm. Chhabhaiya and Bhojane [47] suggested an IoT-enabled internal pipe inspection robot with cloud connectivity for data logging, anomaly notifications, and remote diagnostics. Their approach included live visualization of pipe interiors on cloud dashboards, which is beneficial for utility operators. This connection makes independent robots part of a smart infrastructure network, such that inspection data can be fused with GIS information, maintenance histories, or climatic conditions. Predictive maintenance becomes possible here, as issues are remediated before becoming serious failures. Moreover, robots deployed with environmental probes can send situational context to other network subsystems, and they increase pipeline awareness.

The largest benefit of IoT implementation is operational scale. Inspection bot networks can in real time be coordinated, segmented into routes, and allocate critical areas of attention. Yet such concerns as security for networks, bandwidth constraints, and latency are still obstacles for broad implementation. Standardization and interoperability in hardware platforms, too, present significant challenges. Mills et al. [48] highlighted that modularity in mechanical and communication parts will be essential to achieve plug-and-play robotic units that can be quickly deployed between various pipeline segments. In the meantime, future trends are towards digital twins—virtual copies of entire pipeline systems supplied with real-time robotic data—which can model inspection scenarios and optimize routing and maintenance plans. With infrastructure increasingly being digitized, the cooperation of IoT and in-pipe robotics will be instrumental in turning pipeline monitoring reactive into proactive, promising more efficiency, cost-effectiveness, and eco-friendly protection.

6.4 Materials, Miniaturization, and Roadmaps for Future Research

Advances in materials and miniaturization technologies lead the way to the next-generation development of in-pipe robots. The need for smaller, lighter, more efficient designs arises from the imperative to inspect those portions of pipeline systems previously not accessible because they were too narrow in diameter or presented too much geometrical complexity. Zhang et al. [49] presented a systematic review of worm-like robots and emphasized material-led breakthroughs in flexibility, structural robustness, and locomotion management. Utilization of soft composites, shape-memory alloys, and bioinspired materials is making robots capable of bending, stretching, and conforming like living things. Ren et al. [50] took this conversation further by modeling-based techniques for the prediction of the mechanical response of such robots under different loads and operational states. Their modeling provides insight into how materials act in dynamic, high-friction, or submerged conditions. Such predictive models are essential for pre-testing design variations prior to physical prototyping to save time and cost during development. One of the important trends is the use of multifunctional materials—materials that can be used as structural members as well as sensors or actuators. This obviates the need for large sensor modules and enables robots to be kept compact without compromising functionality. Work is also aimed at energy-harvesting materials that transform motion or environmental heat into energy, minimizing battery usage for extended missions. Verma et al. [46] and Kumar et al. [45] both pointed out that mechanical configuration innovation has to go hand in hand with actuation and material intelligence advancements. Their reviews indicate that robotic pipeline inspection in the future will see a transition from electromechanical systems to biologically inspired, self-adaptive, and even self-healing systems.

In the future, the research world is urging for international cooperation between disciplines—civil engineering, AI, materials science, and robotics—to create standard platforms, open-source test data sets, and field validation methods. These combined efforts will speed up the introduction of dependable, intelligent, and scalable robotic inspection systems that will transform underground infrastructure monitoring.

7. Conclusion

In-pipe robots have become revolutionary machinery in pipeline maintenance and monitoring of infrastructure in most industries, such as water supply, oil and gas, chemical plants, and urban utility. This review has systematically examined the development of robot technologies, including major developments, practical applications, and avenues for future research. The intersection of mechanical innovation, intelligent autonomy, and integrated sensing platforms has transformed the way pipelines are evaluated, making preventive maintenance possible and minimizing risks related to system failure. Section 2 offered a general description of locomotion methods, highlighting the ways in which wheeled, tracked, snake-like, bio-inspired, and hybrid mechanisms all provide unique benefits depending on pipeline geometry, environmental conditions, and operational objectives. As pipelines grow more complex and aged, the need for flexible and adaptive systems that can move through narrow, curved, or submerged pipelines with little human involvement is on the rise. In Section 3, the functional role of smart technologies and AI integration was discussed in detail. Contemporary in-pipe robots are no longer passive searchers but smart agents with localization algorithms, machine learning models for detecting faults, and sensor fusion. These capabilities add accuracy to operations, allow for real-time diagnostics, and minimize reliance on post-processing. AI also aids decision-making, route optimization, and mission-specific adaptations, making these systems vital elements of the larger Industry 4.0 system.

The following sections addressed real-world deployments, technical difficulties, and trends in research. Case studies showed how robots are being utilized today in the field, ranging from teleoperated to completely autonomous units able to navigate over large distances and make high-resolution inspections. At the same time, design and operations challenges like power constraints, sensor degradation, communications failure, and autonomy bottlenecks remain paramount. However, continuous innovations in materials, miniaturization, and hierarchical control systems promise effective solutions for getting past these difficulties. Lastly, the convergence of research trends in Section 6 suggests an interdisciplinary future where robotics, artificial intelligence, IoT, and materials science come together to form intelligent, scalable, and affordable robotic platforms. These will not only

monitor but also predict, adapt, and even repair infrastructure anomalies—changing pipeline maintenance from a reactive to a proactive paradigm. As the world's infrastructure continues to age and grow, the creation and deployment of durable, autonomous in-pipe inspection robots will be critical to sustainable urban development, environmental conservation, and operational resilience. This area, full of innovation, can transform the way we deal with one of the most important yet invisible elements of our contemporary civilization.

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ANNEXURE

Code Snippet A.1 – YOLOv5 Model Training via Google Colab

This script demonstrates the training of a custom YOLOv5m model using Google Colab with GPU support. The dataset containing annotated images of corrosion and cracks was downloaded from Roboflow. The model was trained for 150 epochs and later deployed to Raspberry Pi 5 for live inference.

```
# Step 1: Clone YOLOv5 repository

!git clone https://github.com/ultralytics/yolov5

%cd yolov5


# Step 2: Install YOLOv5 dependencies

%pip install -r requirements.txt


# Step 3: Install Roboflow and download custom dataset

!pip install roboflow

from roboflow import Roboflow

rf = Roboflow(api_key="GACquuNeINudY3ejrUpf")

project = rf.workspace("in-pipe-corrosion").project("in-pipe-corrosion-and-cracks")

version = project.version(4)

dataset = version.download("yolov5")


# Step 4: Train YOLOv5 model

!python train.py --img 640 --batch 16 --epochs 150 \

--data /content/yolov5/in-pipe-corrosion-and-cracks-4/data.yaml \

--weights yolov5m.pt --project /content/yolov5/runs/train \

--name custom_training --hyp data/hyps/hyp.scratch-low.yaml
```

Code Snippet A.2 – Motor Control Using GPIO

The following code snippet controls the direction of the motor using GPIO pins on the Raspberry Pi.

```
if data == "f":  
    GPIO.output(IN1, True)  
    GPIO.output(IN2, False)  
elif data == "s":  
    GPIO.output(IN1, False)  
    GPIO.output(IN2, False)
```

Code Snippet A.3 – YOLOv5 Live Detection Script

This script captures video from the USB camera and runs real-time inference using a custom-trained YOLOv5s model.

Model Load and Video Capture:

```
model = torch.hub.load('ultralytics/yolov5', 'custom', path='best.pt')  
cap = cv2.VideoCapture(0)  
while True:  
    ret, frame = cap.read()  
    results = model(frame)  
    results.print()  
    results.show()
```

Save Detection Results:

```
def write_to_csv(image_name, prediction, confidence):  
    data = {'Image Name': image_name, 'Prediction': prediction, 'Confidence':  
confidence}  
    with open(csv_path, mode='a', newline='') as f:  
        writer = csv.DictWriter(f, fieldnames=data.keys())  
        writer.writerow(data)
```

Draw Bounding Boxes:

```

annotator = Annotator(im0, line_width=line_thickness, example=str(names))

annotator.box_label(xyxy, label, color=colors(c, True))

cv2.imshow(str(p), im0)

```

Code Snippet A.4 – Flask Interface for Detection Result Display

The following code creates a lightweight Flask server to display the YOLOv5 detection output image on a webpage.

```

from flask import Flask, render_template, request

import os

from detect import Start

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('userlog.html')

@app.route('/analyse', methods=['POST'])
def analyse():
    image = request.form['img']
    path = 'static/test/' + image
    Start(path)
    if '.mp4' in image:
        return render_template('userlog.html',

videoDisplay="http://127.0.0.1:5000/static/test/" + image,

videoDisplay1="http://127.0.0.1:5000/static/result/" + image)
    else:

```

```

        return render_template('userlog.html',

ImageDisplay="http://127.0.0.1:5000/static/test/" + image,

ImageDisplay1="http://127.0.0.1:5000/static/result/" + image)

@app.route('/livestream')
def livestream():
    os.system('python livedetect.py')

    return render_template('userlog.html',
videoDisplay1="http://127.0.0.1:5000/static/result/0.mp4")

if __name__ == "__main__":
    app.run(debug=True, use_reloader=False)

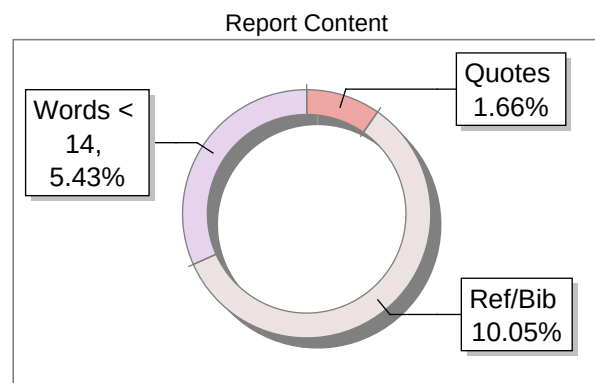
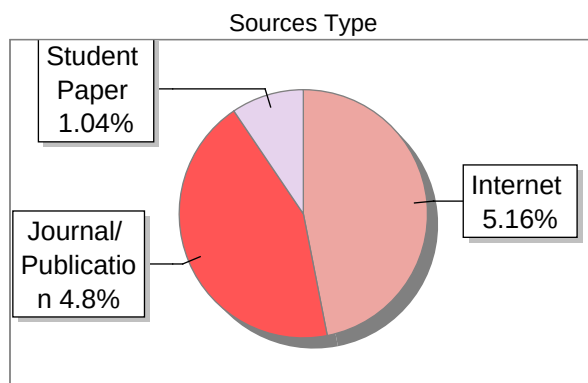
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